

NOTE

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Introduction

Astra RPG Health extends the base framework, [Astra RPG Framework], by adding functionality for managing health and calculating damage for entities. The package is designed with the same design philosophy as the base asset: a Scriptable Object-based architecture to encourage flexibility, modularity, and testability. If you're already familiar with the base package, you'll feel right at home with its features.

You can define your own damage types, designate defensive statistics used to reduce each damage type, configure the damage calculation pipeline with the desired steps, configure lifesteal for certain damage types, define strategies to execute upon entity death, and much more.

Astra RPG Health Vocabulary

Damage Type

Damage types represent the different categories of damage that can be inflicted on entities. Common examples include "Physical", "Magical", "Bleeding", "Drowning", etc. You can create custom damage types to suit your game's needs.

Damage Source

Damage sources represent the origin of the damage inflicted. This is highly specific to your game's context. For example, one game might have damage sources such as "Entity", "Environment", "Potion", etc., while another might want to define more specific damage sources like "Attack", "Spell", "Equipment", "Trap", "Environment", "Damage Over Time", etc. The main difference between damage source and damage type is that damage types categorize damage based on its nature, while damage sources identify where the damage originates. The distinction can be subtle, but it's important for game logic and mechanics. We'll return to these concepts later, where we'll see the practical differences between the two.

Heal Source

Heal sources represent the origin of healing. Similar to damage sources, heal sources are specific to your game's context. Common examples include "Potion", "Spell", "Lifesteal", "Ability", "Environment", etc. In

some games, a classic Heal Source definition might include "Self" and "Ally", as these enable mechanics for increasing healing provided/received based on the caster and target.

Barrier

The concept of temporary HP can take various names across different games, though the underlying mechanic remains the same: provide an amount of extra and ephemeral hit points that are deducted instead of health when damage is taken. In Astra, these temporary hit points are called "Barrier".

Raw and Net Damage

Raw Damage refers to the damage that a certain attack or skill intends to inflict. This damage does not account for defense, critical hits, modifiers, etc. Net Damage is the result of processing Raw Damage while accounting for damage modifiers, damage reduction, barriers, critical hits, etc.

Damage Reduction and Defense Piercing

Primary damage reduction occurs through defensive statistics, which reduce damage based on their value and the assigned damage reduction formula. Each Damage Type can be configured with a specific defensive statistic that will be used to reduce damage of that type. For example, a **Physical Damage** Damage Type could be configured to be reduced by an **Armor** statistic.

Defense Piercing, instead, represents the ability to ignore part or all of the defensive statistic assigned to a Damage Type. Each Damage Type can be configured with a specific piercing statistic that will be used to ignore a portion of the defense when calculating damage. For example, the **Armor Penetration** statistic could be used to determine how much of the **Armor** to ignore when calculating the net damage of an attack that inflicts **Physical Damage**.

Damage Modifiers

Damage modifiers are a low-level abstraction that can alter net damage. They represent either flat or percentage-based modifications to the damage an entity receives. They can be general, applying to all damage received by the entity, or specific to a certain damage type or source.

They are the foundation for implementing a wide range of mechanics — such as elemental resistances and vulnerabilities, status effects, buffs, and debuffs — that affect the damage received based on varying conditions and contexts. Whether a modifier is temporary or permanent depends entirely on the higher-level logic built on top of them. Damage modifiers are generally utilized by the damage calculation pipeline (which we'll see shortly).

Damage Modifiers vs. Defensive Stat-Based Damage Reduction

These are distinct concepts. Stat-based damage reduction (via defensive stats) is the **baseline**: entities are always expected to carry a meaningful value for each defensive stat, and that value **always reduces** incoming damage of the associated type. Damage Modifiers, on the other hand, are backed by stats that

default to zero — meaning they have no effect unless a higher-level system (e.g. a resistance, a weakness, a status effect) changes their value. Additionally, while defensive stats can only ever reduce damage, damage modifiers can go in either direction: **reducing or amplifying** the damage received. Finally, defensive stats often also carry an explicit semantic meaning in the game world. For example, an **Armor** stat clearly indicates a defensive capability that reduces physical damage, and an heavier equipped armor values grants a greater **Armor** value.

Damage modifiers are a more abstract, low-level tool that can be used to implement many different mechanics; they do not carry an inherent, universal meaning. They act as a shared container for damage modifiers coming from various game mechanics. For example, a **Stone Skin** buff, a **Iron Elixir** potion, and a **Barbarian Temper** passive ability, could all rely on the same **Flat Physical Damage Modifier** stat to reduce physical damage. In this case, the **Flat Physical Damage Modifier** stat would be a shared resource that all these mechanics modify to achieve their effect on damage reduction.

The **origin of val values** also differs between the two. Defensive stat values are typically set either as a fixed base value on the entity, or derived from its Class's **GrowthFormula** if classes are used — meaning they scale naturally with the entity's progression. Damage modifier stats, by contrast, usually rely on **flat stat modifiers**: each game mechanic (potion, passive, buff, debuff, etc.) contributes its effect by adding or removing a stat modifier on the underlying stat, temporarily or permanently altering the total modifier applied to incoming damage.

	Defensive Stat Damage Reduction	Damage Modifiers
Default effect	Always active — entities carry a (usually) non-zero defensive stat that continuously reduces incoming damage	Neutral by default — the underlying stat starts at zero and has no effect until explicitly changed
Direction	Can only reduce damage	Can reduce or increase damage
Scope	Specific to a Damage Type — each type can have at most one defensive stat assigned. No reduction exists for Damage Sources	Can target all damage (any type and source), a specific Damage Type , or a specific Damage Source
Nature	Derived from a defensive stat value, processed through a damage reduction formula	Flat or percentage-based modifications driven by higher-level systems (resistances, weaknesses, status effects, etc.)
Duration	Permanent — tied to the entity's stat value	Temporary or permanent, depending on the higher-level systems that apply them

	Defensive Stat Damage Reduction	Damage Modifiers
Value origin	Set as a fixed base value on the entity, or derived from its Class's <code>GrowthFormula</code>	Driven by flat stat modifiers added or removed by individual game mechanics (potions, passives, buffs, debuffs, etc.)

How is Astra RPG Health organized and how does it work?

Astra RPG Health Config

The `AstraRPGHealthConfig` is a `ScriptableObject` that serves as the central configuration point for the Astra RPG Health package. It has several properties that allow you to define how the health and damage systems should behave in your game.

With Astra RPG Framework, no configuration was needed. However, Astra RPG Health needs to be configured to work around the actual instances of the base framework's components defined for your game. For example, if you defined a certain statistic for general damage reduction in your game with Astra RPG Framework, you need to inform Astra RPG Health about it so that it can use it when calculating damage. The needed configuration is kept minimal, and convention-over-configuration is applied where possible to reduce the amount of setup required.

Configuration will be deeply discussed later in [Package Configuration](#).

EntityHealth

`EntityHealth` is a brand new `MonoBehaviour` that you can add to your entities to provide them with health management capabilities. Worth mentioning, it exposes the most important method of the package: `TakeDamage()`, which allows you to inflict damage on the entity.

Damage Type

`DamageType` is a `ScriptableObject` that represents a specific type of damage. Each damage type can be optionally configured with a defensive `Stat` that will be used to reduce incoming damage of that type. If a defensive stat is assigned, also a piercing stat can be assigned to ignore a portion of the defense when calculating damage.

Both for the defensive and piercing stat, you can select a `DamageReductionFormula` and a `DefenseReductionFormula` respectively, to define how the stats will affect damage reduction and defense piercing.

Damage Source

`DamageSource`, derived from `ScriptableObject`, represents the origin of the damage inflicted. They don't have any specific properties, but they can be assigned to other objects of the package to create specific behaviors based on the damage source. We will see for what and how in the Workflows.



Damage Reduction Functions

The package comes with three built-in `DamageReductionFunctions` that you can use to define how defensive stats reduce incoming damage:

-  `FlatDamageReductionFn`: Reduces damage by a flat amount based on the defense stat value.
-  `PercentageDamageReductionFn`: Reduces damage by a percentage based on the defense stat value.
-  `LogarithmicDamageReductionFn`: Reduces damage using a logarithmic scale based on the defense stat value.

In case of need, custom damage reduction functions can be created by extending the `DamageReductionFn` class.



Defense Piercing Functions

As for damage reduction functions, the package comes with three built-in `DefensePiercingFunctions` that you can use to define how piercing stats ignore a portion of the defense stat:

-  `FlatDefensePiercingFn`: Ignores a flat amount of the defense stat based on the piercing stat value.
-  `PercentageDefensePiercingFn`: Ignores a percentage of the defense stat based on the piercing stat value.
-  `LogarithmicDefensePiercingFn`: Ignores a portion of the defense stat using a logarithmic scale based on the piercing stat value.

Also in this case, custom defense piercing functions can be created by extending the `DefensePiercingFn` class.



Heal Source

`HealSource`, derived from `ScriptableObject`, represents the origin of healing. Similar to `DamageSource`, they don't have any specific properties, but they can be assigned to other objects of the package to create specific behaviors based on the heal source. We will see for what and how in the Workflows.



Damage Calculation Strategy

The damage calculation pipeline is the component of the framework responsible for processing raw damage and producing net damage. The pipeline will use a given `DamageCalculationStrategy` to determine the sequence of steps to apply when calculating damage.

Therefore, a `DamageCalculationStrategy` is a `ScriptableObject` that defines a sequence of `DamageSteps` to be executed in order when calculating damage.

On-death & on-resurrection **GameActions**

Astra RPG Framework v1.4.0 introduced the concept of **GameAction**: a modular and reusable unit of game logic that can be assigned via the inspector to various components to define custom behaviors.

Astra RPG Health leverages Game Actions to define custom behaviors when an entity dies or is resurrected. The framework allows you to assign a default on-death Game Action that will be executed when any entity dies, unless the entity has its own custom on-death Game Action assigned. Similarly, a default on-resurrection Game Action can be assigned to be executed when an entity is resurrected.

Lifesteal Configuration

LifestealConfig, deriving from **ScriptableObject**, allows you to define how lifesteal mechanics work for specific damage types. This allows to bind a statistic to each damage type you want to have lifesteal for. That statistic will be used to calculate the amount of health to restore to the attacker when they deal damage of that type.

The configuration allows also to configure the damage pipeline timing of the lifesteal effect. For example, you might want lifesteal to occur before or after damage reduction is applied. We will see this in detail later in the Workflows.

Health Scaling Component

Astra RPG Health provides a brand new **HealthScalingComponent** that you can use in your **ScalingFormulas** to have skills or abilities scale based on either the attacker or the target's health. You can choose to scale upon one or more among Maximum HP, Current HP, and Missing HP.

Experience Collector

An **ExpCollector** is a **MonoBehaviour** that you can add to your entities to allow them to collect experience points from entities that die and have an  **ExpSource** component attached. The **ExpCollector** can be configured with different strategies ( **ExpCollectionStrategy**) to define under which conditions experience is collected from the dead entity.

Experience Collection Strategy

An **ExpCollectionStrategy** is a **ScriptableObject** that defines the logic for determining if an entity collects experience upon the death of another entity.

More Game Events

Astra RPG Health comes with many new Game Events that you can use to react to health&damage-related events in your game. Some of the most important ones are:

- **PreDamageGameEvent**: Triggered before damage is applied to an entity. Useful for modifying or canceling damage. Use this for implementing custom passives or effects that need to react before damage is taken.
- **DamageResolutionGameEvent**: Triggered when an entity takes damage. Can be used to react to damage being applied.
- **EntityDiedGameEvent**: Triggered when an entity dies.
- **EntityHealedGameEvent**: Triggered when an entity is healed.
- **EntityResurrectedGameEvent**: Triggered when an entity is resurrected.

And many more events. We will discuss them in detail later in the Workflows.

Package Configuration

Astra RPG Framework needed no configuration. The health package, however, needs to be configured to have the system work around the specific instances of your game. For example, if you defined a "Super Duper All-damage resistance" `Stat` in your game, you need to tell Astra RPG Health to use it when calculating damage.

The package uses a flexible configuration system that balances convenience with explicit control. This page explains how to set up and configure the health system for your game.

Configuration Overview

The Astra RPG Health system uses a **two-tier configuration architecture**:

1. **Global Settings** (`AstraRpgHealthGlobalSettings`) - A lightweight pointer stored in `Resources` that references your active configuration
2. **Gameplay Configuration** (`AstraRpgHealthConfigSO`) - The actual configuration containing all gameplay parameters

This separation allows you to:

- Switch between different configuration profiles easily (e.g., for testing, different game modes)
- Keep configuration data separate from the loading mechanism
- Support convention-based fallbacks for quick prototyping

Global Settings

Automatic Setup

The package automatically creates the Global Settings asset on first import or when the editor loads. You don't need to do anything manually.

What happens automatically:

1. The `AstraRpgHealthGlobalSettings.asset` is created in `Assets/Resources/`
2. If a default configuration exists (e.g., from imported samples), it's automatically assigned
3. The system is immediately ready to use

NOTE

Default location: `Assets/Resources/AstraRpgHealthGlobalSettings.asset`

Project Settings

You can manage the package configuration through Unity's Project Settings window:

1. Open **Edit** → **Project Settings**
2. Navigate to **Astra RPG Health**
3. Assign your desired **Active Config Profile**

Status Indicators:

-  **Gray "Using Explicit Configuration"** - A configuration is explicitly assigned
-  **Yellow "Using Fallback"** - No explicit configuration; using convention-based fallback
-  **Red "No Configuration Found"** - Critical: No configuration available

Quick Actions:

- **Create New Config Asset** - Opens a save dialog to create a new configuration

Manual Configuration (alternative to Project Settings)

You can also manage the package configuration manually by directly editing the `AstraRpgHealthGlobalSettings` asset in the `Assets/Resources` folder. This approach allows you to assign a specific `AstraRpgHealthConfigSO` asset without using the Project Settings window.

Convention Over Configuration

The package follows a **convention-over-configuration** philosophy to reduce setup friction:

Fallback Resolution

If no explicit configuration is assigned in Project Settings, the system automatically searches for a configuration named:

`Astra Rpg Health Config`

located in any `Resources` folder in your project.

Search Order

The configuration provider uses a **three-step loading strategy**:

1. **Explicit Configuration** (Project Settings)
 - Loads `AstraRpgHealthGlobalSettings` from `Resources/AstraRpgHealthGlobalSettings`
 - If it has an `ActiveConfig` assigned, use it
2. **Convention-Based Fallback**

- Searches for **Astra Rpg Health Config** in any **Resources** folder
- Logs a warning indicating fallback usage

3. Error State

- If neither is found, logs an error with instructions
- System will not function until a configuration is provided

TIP

For production projects, always use **explicit configuration** via Project Settings for clarity and control.

Configuration Loading Strategy

The health configuration is loaded lazily on first access and cached for performance:

```
// Automatically loads configuration on first access
var config = AstraRpgHealthConfigProvider.Instance;

// Pre-load during initialization to avoid runtime overhead
AstraRpgHealthConfigProvider.WarmUp();

// Force reload (useful for testing)
AstraRpgHealthConfigProvider.Reset();
```

When is the configuration loaded?

- Automatically before the first scene loads (via **RuntimeInitializeOnLoadMethod**)
- Lazily when first accessed via **AstraRpgHealthConfigProvider.Instance**
- Explicitly when calling **WarmUp()**

Creating Configuration Assets

If for any reason you need to create a new **AstraRpgHealthConfigSO** asset, you can do so via two methods:

Via Project Settings

1. Open **Edit** → **Project Settings** → **Astra RPG Health**
2. Unassign any existing configuration, if any
3. Click **Create New Config Asset**

4. Choose a save location
5. The new configuration is automatically assigned

Via Asset Menu

1. Right-click in the **Project Window**
2. Select **Create** → **Astra RPG Health** → **Configuration**
3. Name your configuration
4. Assign it in Project Settings or in the Global Settings asset

The red asterisks (*) indicate fields that are required for the system to function properly. Make sure to assign them before entering Play Mode to avoid runtime errors.

Health Configuration Reference

NOTE

In the `AstraRpgHealthConfigSO` asset, you can hover over each field to see a tooltip with a brief description.

The `AstraRpgHealthConfigSO` asset contains all gameplay parameters for the health system. Below is a detailed explanation of each field.

Health

Health Attributes Scaling

Type: `AttributesScalingComponent`

Required: No

Description: Defines how entities' maximum health scales based on character attributes (e.g., Vitality, Endurance).

See also: [Astra RPG Framework Scaling documentation](#) 

Generic Flat Heal Amount Modifier Stat

Type: `Stat`

Required: No

Description: The stat that modifies **all** healing received by an entity as a flat amount.

Stacking Behavior:

- Combines **additively** with source-based flat amount modifications

How it works:

- The stat value represents a **flat amount modifier**
- Positive values increase healing, negative values decrease it
- Example: A value of 25 means +25 extra HP healing received

Example:

- Base Heal: 100 HP
- Generic Heal Amount Modifier: 25 (means +25 HP healing received)
- Final Heal: $100 + 25 = 125$ HP

Generic Percentage Heal Amount Modifier Stat

Type: Stat **Required:** No **Description:** The stat that modifies **all** healing received by an entity as a percentage. It is applied on top of the flat heal amount modifiers.

Stacking Behavior:

- Combines **additively** with source-based percentage modifications

How it works:

- The stat value represents a **percentage modifier**
- Positive values increase healing, negative values decrease it
- Example: A value of 20 means +20% extra healing received

Example:

- Base Heal: 100 HP
- Generic Flat Heal Amount Modifier: 20 (means +20 extra HP healing received)
- Generic Percentage Heal Amount Modifier: 20 (means +20% healing received)
- Final Heal: $(100 + 20) \times 1.20 = 144$ HP

WARNING

Remember to remove the default lower bound of 0 from your flat and percentage heal modifiers if you want to allow negative values for healing reduction effects. By default, the base framework sets a lower bound of 0 on all stats.

Damage

Default Damage Calculation Strategy

Type: `DamageCalculationStrategy`

Required: No

Description: The default strategy used to calculate net damage when an entity doesn't specify its own strategy.

The default strategy applied to entities that do not have a custom one assigned in their `EntityHealth` component. In most cases a single global default is sufficient; override it per-entity when a different calculation pipeline is required. See [Damage Calculation Strategy](#) for details on how strategies work.

Generic Flat Damage Modification Stat

Type: `Stat`

Required: No

Description: A universal flat damage modifier that applies to **all damage received**, regardless of type or source.

Applied by `ApplyFlatDmgModifiersStep` when that step is included in the active strategy.

Stacking Behavior:

- Combines **additively** with Type and Source **flat** modifications

Example:

- Incoming Damage: 150
- Generic Flat Damage Modification: -20 (means -20 HP of damage taken)
- **Damage Reduction:** -20 → Final Damage: **130**

Generic Percentage Damage Modification Stat

Type: `Stat` **Required:** No **Description:** A universal percentage damage modifier that applies to **all damage received**, regardless of type or source.

Applied by `ApplyPercentageDmgModifiersStep` when that step is included in the active strategy.

Stacking Behavior:

- Combines **additively** with Type and Source **percentage** modifications

Example:

- Incoming Damage: 150
- Generic Percentage Damage Modification: -20 (means -20% damage taken)
- **Damage Reduction:** -20% → Final Damage: $150 \times 0.80 =$ **120**

NOTE

If the entity to be healed doesn't have the specified flat/percentage heal modification stats, they will be considered as having a value of 0 for those stats, and therefore no modification will be applied to the healing amount for that entity.

Health Regeneration

Health Regeneration Source

Type: `HealSourceSO`

Required: No

Description: The heal source used for passive regeneration effects.

Use cases:

- Allows tracking and modifying regeneration separately from active healing
- Can be used for effects like "Increase Regeneration by 50%"
- Tracking passive healing in analytics or combat logs

Passive Health Regeneration Stat (HP/10s)

Type: `Stat`

Required: No

Description: Determines the amount of health regenerated passively.

WARNING

The stat value represents health regenerated **per 10 seconds**.

Calculation:

`Health Per Tick = (Stat Value / 10) * Interval In Seconds`

Example:

- Stat Value: 50 HP/10s
- Interval: 1 second
- Health Per Tick: $(50 / 10) * 1 = 5 \text{ HP per second}$

Passive Health Regeneration Interval

Type: `float`

Default: 1.0 seconds

Required: Yes (must be > 0)

Description: The time (in seconds) between passive regeneration ticks.

Configuration Examples:

Interval	Stat Value	Result
1.0s	50 HP/10s	5 HP every 1 second
0.5s	50 HP/10s	2.5 HP every 0.5 seconds
2.0s	50 HP/10s	10 HP every 2 seconds

⚠ **WARNING**

Smaller intervals increase CPU overhead. Recommended range: 0.5s - 2.0s.

Suppress Passive Regeneration Events

Type: `bool`

Required: No

Description: If enabled, prevents triggering any heal events during passive regeneration ticks. Keep it disabled if you need to track passive regeneration in a combat log or if you have effects that trigger on heal events and should also apply to passive regeneration. If your game doesn't require any of the above and you want to minimize overhead, you can enable this option and skip all heal-related logic during regeneration ticks. Useful if your game has a lot of entities with passive regeneration and you want to optimize performance. If you need both to keep sending regeneration events and to minimize overhead, I advise to increase the regeneration interval and to keep the "Suppress Passive Regeneration Events" option disabled. This way you can have less regeneration ticks and still trigger events for each tick.

Manual Health Regeneration Stat

Type: `Stat`

Required: No

Description: Determines health regenerated when triggering manual regeneration via API. The amount of health regenerated is equal to the value of this stat.

Use Cases:

- **Turn-based systems:** Regenerate health at the end of each turn
- **Rest mechanics:** Trigger regeneration when resting at campfires
- **Time-skip systems:** Apply regeneration for elapsed time

API Usage:

```
// Trigger manual regeneration  
entityHealth.ManualHealthRegenerationTick();
```

Lifesteal

Lifesteal Config

Type: `LifestealConfigSO`

Required: No

Description: Configuration for lifesteal mechanics (healing based on damage dealt).

Typical Settings:

- Lifesteal percentage stat
- Heal source for lifesteal effects
- Restrictions (e.g., only on critical hits, only physical damage)

See also: [Lifesteal](#)

Experience

Default Exp Collection Strategy

Type: `ExpCollectionStrategySO`

Required: No

Description: The fallback strategy used by any `ExpCollector` that does not have a custom strategy assigned on the component itself.

Setting this once is usually all you need: every collector in the project that relies on the default will automatically use it without requiring per-entity configuration. Override it on individual `ExpCollector` components only when a specific entity needs different award logic.

See also: [Experience Collection](#)

Death

Default On Death Game Action *

Type: `GameAction`

Required: Yes

Description: The game action executed when an entity dies (if the entity doesn't have its own on-death game action). Use a composite game action to chain multiple effects.

Common on-death Game Action Ideas:

- **Destroy GameObject** - Removes the entity from the scene
- **Ragdoll** - Enables ragdoll physics
- **Respawn** - Respawns the entity after a delay
- **Loot Drop** - Spawns loot and destroys the entity

Example:

Death → `Execute` composite `on-death` Game Action → [Spawn Death VFX → `Drop` Loot → Destroy GameObject]

Default On Resurrection Game Action

Type: `GameAction`

Required: No

Description: The game action executed when an entity is resurrected (if the entity doesn't have its own on-resurrection game action). Use a composite game action to chain multiple effects.

Common on-resurrection Game Action Ideas:

- **Simple Resurrection** - Restores health and enables the entity
- **Resurrection VFX** - Plays visual effects during resurrection
- **Stat Penalties** - Applies temporary debuffs after resurrection

Default Resurrection Source *

Type: `HealSourceSO`

Required: Yes

Description: The heal source used when an entity is resurrected via convenience overload `Resurrect` methods, or via the built-in [Resurrect Game Action](#).

Use cases:

- Categorizes resurrection healing separately from normal healing
- Allows effects like "Increase Resurrection Healing by 50%"
- Used for analytics and gameplay feedback

Global Events

Global Events are **GameEvent** ScriptableObjects that broadcast health-related information to the **entire game**. They are assigned once in the configuration asset and shared by all entities, ensuring a single authoritative source for framework-level communication.

⚠ WARNING

Three events are **required** — the framework logs an assertion error at startup if they are missing. Assign them before entering Play Mode.

Global Pre Damage Info Event *

Type: **PreDamageGameEvent**

Required: Yes

Description: Raised before any entity takes damage and before the [damage calculation pipeline](#) runs. Carries the full **PreDamageContext** (raw amount, type, source, dealer, critical state). Subscribe to this event to implement effects that inspect or modify incoming damage, such as damage-amplifying debuffs or conditional immunity logic.

Global Damage Resolution Event *

Type: **DamageResolutionGameEvent**

Required: Yes

Description: Raised after any entity takes or ignores damage. Carries the full **DamageResolutionContext** including the outcome (**Applied** or **Prevented**), net amount, and prevention reasons. Also used internally by the framework for lifesteal resolution.

Global Entity Died Event *

Type: **EntityDiedGameEvent**

Required: Yes

Description: Raised when an entity's HP reaches the death threshold. Carries **EntityDiedContext** with the entity reference and the damage context that caused the death. Used by the [Experience Collection](#) system.

Global Max Health Changed Event

Type: **EntityMaxHealthChangedGameEvent**

Required: No

Description: Raised when any entity's total max HP changes (both increases and decreases). Carries **EntityMaxHealthChangedContext**. Useful for updating UI, recalculating derived stats, or triggering game-wide reactions to stat changes.

Global Gained Health Event

Type: `EntityGainedHealthGameEvent`

Required: No

Description: Raised whenever any entity gains HP (from healing, resurrection, or max HP modifier increases). Carries `EntityHealthChangedContext`.

Global Lost Health Event

Type: `EntityLostHealthGameEvent`

Required: No

Description: Raised whenever any entity loses HP (from damage, or max HP modifier decreases). Carries `EntityHealthChangedContext`.

Global Pre Heal Event

Type: `PreHealGameEvent`

Required: No

Description: Raised before any entity is healed and before healing modifiers are applied. Carries `PreHealContext` (base amount, source, healer, target). Subscribe to modify heal amounts or trigger pre-heal effects.

Global Entity Healed Event

Type: `EntityHealedGameEvent`

Required: No

Description: Raised after any entity is healed. Carries `ReceivedHealContext` with the final net heal amount after all modifiers.

Global Entity Resurrected Event

Type: `EntityResurrectedGameEvent`

Required: No

Description: Raised after any entity is resurrected. Carries `ResurrectionContext` with HP before and after resurrection.

Troubleshooting

"No Configuration found!" error

Cause: No configuration is assigned and no fallback exists.

Solution:

1. Check **Project Settings** → **Astra RPG Health**

2. Assign a configuration or create a new one
3. Alternatively, create a config named **Astra Rpg Health Config** in a **Resources** folder

"Using Fallback" warning

Cause: No explicit configuration assigned in Project Settings.

Solution:

1. Open **Project Settings** → **Astra RPG Health**
2. Assign the fallback configuration explicitly
3. This warning is just informational and won't break functionality

Configuration not updating in Play Mode

Cause: Configuration is cached on first access.

Solution:

```
// Force reload  
AstraRpgHealthConfigProvider.Reset();
```

Missing Resources folder

Cause: The **Assets/Resources/** folder doesn't exist.

Solution: The package creates it automatically. If it's missing, create it manually:

1. Right-click **Assets**
2. Create → Folder → Name it **Resources**
3. Restart Unity to trigger the bootstrapper

EntityHealth Component

With Astra RPG Health, the new **EntityHealth** MonoBehaviour allows you to add health points to an entity. In this way, the entity can take damage and, consequently, die.

Here is an example of the **EntityHealth** component:

The screenshot shows the configuration for the **EntityHealth** component on a **Player** entity. The component is selected, and its settings are visible. The **Health** section is expanded, showing various health-related properties. The **Damage** section is also expanded, showing damage calculation settings. The **Death** section is expanded, showing death-related settings. The **Events** section is collapsed.

Property	Value
Script	EntityHealth
Health	
Use Class Max HP	☐
Base Max HP	☒ Use Constant 100
Total Max HP	☒ Use Constant 100
Current HP	☒ Use Constant 100
Barrier	☒ Use Constant 0
Passive Health Regeneration	☐
Restore HP On Level Up	☒
Damage	
Custom Damage Calculation Strategy	None (Damage Calculation Strategy)
Override Damage Calculation Strategy	None (Damage Calculation Strategy)
Is Immune	☐
Death	
Health Can Be Negative	☐
Override On Death Game Action	None (Game Action `1)
Override On Resurrection Game Action	None (Game Action `1)
Events	

The events section is collapsed by default since there are many events, and it is more practical to expand it only when necessary.

Let's proceed in order and analyze every property of the **EntityHealth** component.

Health

- **Use Class Max HP:** Boolean indicating whether to use the max health points defined in the entity's class as the base max health. If disabled, you can manually specify the base max health for the entity.
- **Base Max HP:** LongRef representing the **base** max health of the entity. If "Use Class Max HP" is enabled, this value is marked with teal **RO** (Read Only). Read Only, for LongRef fields, makes **const** values non-editable, and suggests not manually modifying values contained by the associated LongVar variable, if a **const** value is not used.
- **Total Max HP:** LongRef representing the **total** max health of the entity, calculated based on the base max health and any modifiers. This field is always read-only (RO) and is automatically updated when base max health or modifiers change.
- **Current HP:** LongRef representing the entity's current health points. Editable field that is automatically updated when the entity takes damage or is healed. This field is also updated if **Base Max HP** is modified from the inspector.

NOTE

If **Current HP** drops to 0, modifying **Base Max HP** will not update **Current HP**. This is intended behavior, as if an entity is dead, it makes no sense for its health points to be modified by a change in max health. If instead the entity is alive, modifying **Base Max HP** will update **Current HP** accordingly even if **Current HP** is currently 0. This can happen if, for example, you change from 10 base max HP to 25 by deleting the text box with backspace: 10 (backspace) -> 1 (backspace) -> 0 (2 pressed) -> 2 (5 pressed) -> 25. Note that by deleting all numbers and leaving the box empty, **Base Max HP** becomes 0, and consequently **Current HP** also becomes 0. At this point, the entity is considered dead.

- **Barrier:** LongRef representing any barrier points (or temporary HP) of the entity. The barrier absorbs damage before it affects the entity's health points.
- **Passive Health Regeneration:** Boolean that decrees whether the entity passively regenerates HP over time or not. The regeneration frequency, as well as the statistic to consider for the amount of passively regenerated HP, are defined in the [Astra RPG Health Config](#) configuration.
- **Restore HP On Level Up:** Boolean indicating whether the entity is fully healed when leveling up.

Damage

For a better understanding of the first two properties of this section, I recommend taking a look at the [Damage Calculation Strategy](#) documentation. Simply put, a Damage Calculation Strategy defines how the damage an entity is about to take is calculated (e.g., applying damage reduction for the defensive stat first, or damage absorption by the barrier first, when to apply the critical multiplier, etc.). Also recall that a default strategy can be assigned via configuration. See [Default Damage Calculation Strategy](#).

- **Custom Damage Calculation Strategy:** Field of type `DamageCalculationStrategy`. If the entity should use a custom damage calculation strategy, you can specify it here. This, if defined, takes precedence over the default one defined via configuration. A common use case could be, for example, a boss that cannot take more than 10% of its max health in damage at a time.
- **Override Damage Calculation Strategy:** Field of type `DamageCalculationStrategy`. If defined, it takes precedence over all other damage calculation strategies. Designed to be assigned at runtime to implement special effects or for testing/debug. For example, an entity is affected by a debuff that turns all physical damage into guaranteed critical hits. This debuff could therefore be implemented through a custom strategy assigned to the entity in this field.
- **Is Immune:** Boolean indicating whether the entity is immune to all damage or not. If enabled, the entity will take no damage, regardless of the damage calculation strategy used.

Death

- **Health Can Be Negative:** Boolean indicating whether the entity's health points can drop below 0 before dying. If disabled, the entity's health points will never drop below 0, and the entity will die as soon as its health points reach 0. If enabled, the entity's health points can drop below 0, and the entity will die only when its health points reach a specified negative value (Death Threshold).
- **Death Threshold:** LongRef representing the entity's death threshold. Visible only if "Health Can Be Negative" is enabled. If the entity's health points reach this threshold, the entity dies.
- **Override On Death Game Action:** [Game Action](#) that is automatically executed when the entity dies. If defined, this takes precedence over the one defined [in the configuration](#). Useful for implementing special behaviors upon an entity's death (for example, entities that explode upon death dealing area damage).
- **Override On Resurrection Game Action:** [Game Action](#) that is automatically executed when the entity is resurrected. If defined, this takes precedence over the one defined [in the configuration](#). Useful for implementing special behaviors upon an entity's resurrection.

Damage vs RemoveHealth and Heal vs AddHealth

`EntityHealth` provides a couple of public methods to increase current HP and two to decrease them:

- `Heal` and `AddHealth` to increase current HP.
- `TakeDamage` and `RemoveHealth` to decrease current HP.

`TakeDamage` and `Heal` are extensively documented in the [Dealing Damage to an Entity](#) and [Healing an Entity](#) sections respectively. However, this sounds like a good moment to introduce the difference between these two pairs of methods, and when to use one or the other.

It is important to clarify why two different methods exist to increase and decrease current HP, and when to use one or the other.

`AddHealth` and `RemoveHealth` operate at a lower level of abstraction than `Heal` and `Damage`, as they directly modify the entity's current HP by a specified `long` value, without passing through the damage calculation pipeline or without taking into account any healing or damage modifiers. They are very predictable and direct methods, and are mainly used internally by the framework. These methods can be useful in specific situations where the gain of HP is not due to healing, or the loss of HP is not due to damage, but rather to particular mechanics that require a direct modification of current HP. For example, suppose that following a cutscene we want to force the player's HP to 1. In this case, we could use `RemoveHealth` to remove all HP except 1, without having to worry about any damage modifiers or the damage calculation strategy that would alter the damage taken based on the player's stats and equipment, as well as raising damage events that could trigger game logic we don't want to activate in this specific case.

`Heal` and `TakeDamage`, on the other hand, are more complex methods that take into account various factors such as the damage calculation strategy, damage immunity, healing modifiers, etc. These methods are intended to be used primarily by game developers to apply damage and healing to entities, as they guarantee that all mechanics and rules of the health system are respected. For example, if you want to inflict damage on an entity, it is advisable to use the `TakeDamage` method, so that the damage is calculated correctly based on the configured damage calculation strategy, and that any immunities or modifiers are taken into consideration. Unsurprisingly, `Heal` and `TakeDamage` internally use `AddHealth` and `RemoveHealth` to effectively modify the entity's current HP after calculating the net HP gain or loss to apply.

In 99% of cases, you will use `Heal` and `TakeDamage` to apply healing and damage to entities. `AddHealth` and `RemoveHealth` are available to cover rarer and more particular use cases.

Events

Events are, by default, collapsed as there are many of them and they would expand excessively in the inspector. By opening them, we should see something like this:

▼ Events



Global events are configured in the `AstraRpgHealthConfig` asset (shared across all entities). Only entity-specific extra events are defined here.

Pre-Damage Events

+ Add Extra Pre Damage Event

Damage Resolution Events

+ Add Extra Damage Resolution Event

Health Change Events

+ Add Extra Gained Health Event

+ Add Extra Lost Health Event

+ Add Extra Max Health Changed Event

Death Events

+ Add Extra Entity Died Event

Heal Events

+ Add Extra Pre Heal Event

+ Add Extra Entity Healed Event

Resurrection Events

+ Add Extra Entity Resurrected Event

In the image, you don't see any assigned event. Therefore, this is the view of the inspector you should get when you freshly add the `EntityHealth` component to an entity.

Through `EntityHealth` we can assign *Extra Events*. However, for each event type (e.g., Pre Damage Info Event, Entity Died Event, etc.) you should know that there are two categories: Global Events and Extra Events.

Global Events

Global Events broadcast health-related information to the entire game and are configured once at the project level in the [AstraRpgHealthConfig](#) asset — not per entity on `EntityHealth`. This guarantees a single authoritative source for framework-level communication and eliminates the risk of misconfiguration caused by different entities referencing different event assets.

The framework uses several global events internally: for example, the *Damage Resolution* event powers *lifesteal*, and the *Entity Died* event is what experience collectors listen to for XP awards. See [Global Events](#)

in Package Configuration for the full reference, including which events are required and which framework features depend on each.

Extra Events

Extra Events, instead, are designed to transmit information to specific components, to a single entity, or to a restricted circle of entities. A practical example is the communication between the player's EntityHealth and the HUD displaying their HP: only the player possesses a dedicated HUD, so it is useful to assign an exclusive event for this interaction. In this way, the GameEvent associated with the player's EntityHealth is listened to only by the HUD, guaranteeing clear compartmentation and greater efficiency. Thus, the HUD does not receive events from all entities, but only those relevant to the player.

These events are assigned directly on the **EntityHealth** component. For each event type, you can assign multiple Extra Events, which will be raised simultaneously along with the corresponding Global Event when the relevant action occurs (e.g., the entity takes damage, the entity dies, etc.).

Events Breakdown

Here is a detailed description of each event. Each type of event has both the global event and a list of extra events, but it is sufficient to describe each event once.

Damage Related Events

To better understand the first two events of this section, I recommend taking a look at the [Damage](#) documentation to get a better understanding of damage in Astra RPG Health.

- **Pre Damage Info Event:** Event raised before the entity takes damage, and before the [damage calculation pipeline](#) calculates the net damage. This event transmits information about the damage the entity is about to take, such as the damage type, the damage source (an entity, **null** if not applicable), the damage source type (e.g., environmental, skill, trap, etc.), the raw damage you intend to inflict, and other relevant information. For more details regarding the context parameter and its fields, refer to the API documentation [PreDamageContext](#). This event can be useful for implementing passive abilities or, in general, advanced mechanics that trigger effects in response to specific conditions related to the damage an entity is about to take. For example, a debuff or status modifier that amplifies critical damage taken by 50% when the damage type is "Fire", or a passive ability that negates all instances of damage currently being taken if they had raw damage less than 50.
- **Damage Resolution Event:** Event raised after the entity has taken or ignored damage. Similarly to the context parameter of the previous event, this event transmits detailed information about the damage just taken or ignored. For more details regarding the context parameter and its fields, refer to the API documentation [DamageResolutionContext](#).

In general, the thumb rule for deciding whether to use the Pre Damage Info Event or the Damage Resolution Event is the following: if you want to manipulate the damage an entity is about to take, it is

better to subscribe to Pre Damage Info Event, and modify the context as desired; if instead you want to react to the damage an entity has just taken, and trigger effects in response to it, it is better to subscribe to Damage Resolution Event, and react based on the information transmitted by its context.

Health Related Events

- **Gained Health Event:** Event raised when the entity gains HP, whether through healing or other mechanisms (e.g., Max HP modifiers that caused a gain of health). Refer to the [EntityHealthChangedContext](#) API for more details on the context parameter.
- **Lost Health Event:** Event raised when the entity loses HP, whether through taking damage or other mechanisms (e.g., Max HP modifiers that caused a loss of health). The context parameter of this event is the same as the previous one, so refer to the [EntityHealthChangedContext](#) API for more details.

NOTE

Because `TakeDamage` and `Heal` internally invoke `RemoveHealth` and `AddHealth`, calling these methods will also raise the `EntityGainedHealth` and `EntityLostHealth` events respectively.

- **Max Health Changed Event:** Event raised when an entity's total max health points change. This event is raised both when total max hp increase and when they decrease. See [EntityMaxHealthChangedContext](#) API for more details on the context parameter.
- **Entity Died Event:** Event raised when the entity dies. See [EntityDiedContext](#) API for more details on the context parameter.

Healing Related Events

- **Pre Heal Event:** Event raised before the entity is healed. This event transmits information about the healing the entity is about to receive, such as the amount of HP you intend to heal, the entity that provided the healing (`null` if not applicable), the heal source (e.g., skill, item, potion, etc.) and other relevant information. For more details regarding the context parameter and its fields, refer to the API documentation [PreHealContext](#). The amount of health points intended to heal, transmitted by this event, is the value before applying any healing modifiers. This event can be useful for implementing passive abilities or, in general, advanced mechanics that trigger effects in response to specific conditions related to the healing an entity is about to receive. For example, a buff that amplifies healing derived from potions (heal source) by 30%.
- **Entity Healed Event:** Event raised when the entity has been healed. This event transmits information about the healing just received, such as the amount of HP the entity actually gained after the application of any healing modifiers. For more on the context parameter see the [Received HealContext](#) API.

Here too, as for damage events, the thumb rule for deciding whether to use the Pre Heal Event or the Entity Healed Event is the following: if you want to manipulate the healing an entity is about to receive, it is better to subscribe to Pre Heal Event, and modify the context as desired; if instead you want to react to the healing an entity has just received, and trigger effects in response to it, it is better to subscribe to Entity Healed Event, and react based on the information transmitted by its context.

Resurrection Related Events

- **Entity Resurrected Event:** Event raised when the entity is resurrected. See [ResurrectionContext](#) API for more details on the context parameter.

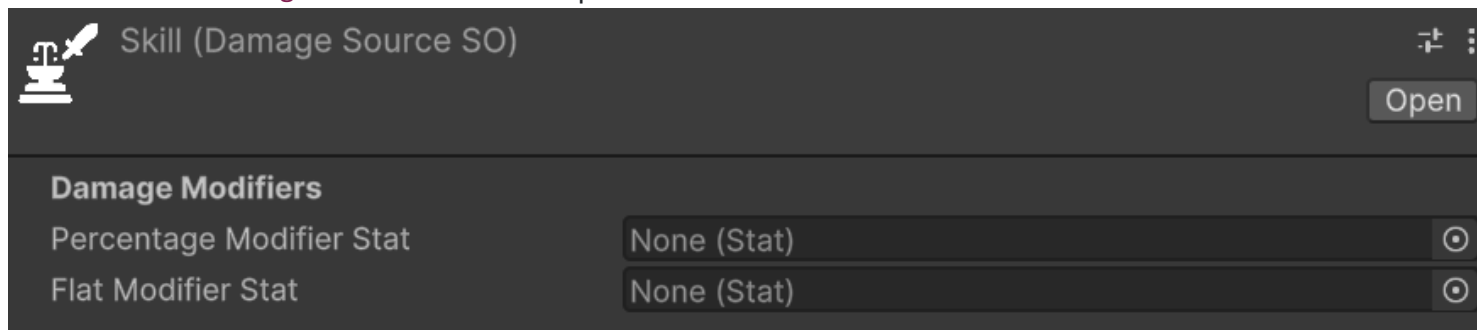
Damage

Damage Sources

Relative path: `Damage Source`

A `DamageSourceSO` represents the source of the damage. Some examples of `DamageSourceSO` could be: skill, base attack, fall damage, trap, environmental, etc. The `DamageSourceSO` is used to categorize the damage and can be used in various mechanics, such as damage modifiers that only apply to specific damage sources, for triggering specific effects when taking damage from a certain source, or for tracking damage statistics based on the source.

An instance of `DamageSourceSO`, in the inspector, should look like this:



Damage Sources - Modifiers

There are two properties to set in a `DamageSourceSO`:

- **Percentage Modifier Stat:** the statistic to consider in an entity to apply percentage, specific damage modifiers for this damage source. Positive values of this statistic increase the damage received from this source, while negative values decrease it.
- **Flat Modifier Stat:** the statistic to consider in an entity to apply flat, specific damage modifiers for this damage source. Positive values of this statistic increase the damage received from this source, while negative values decrease it.

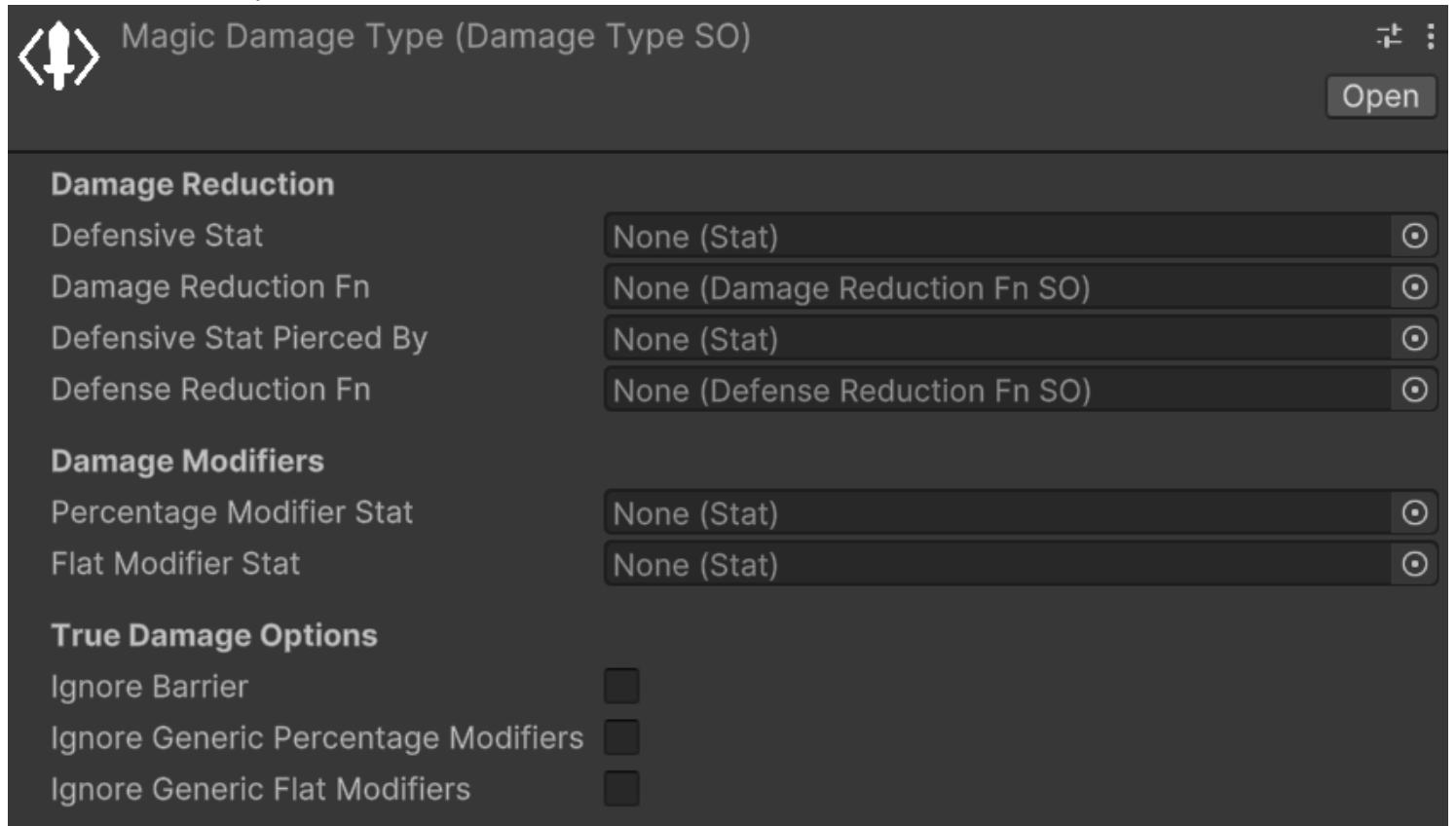
⚠ WARNING

If the entity lacks a percentage or flat damage source modifier statistic, an error will be logged when applying damage from that source. Ensure all entities with an `EntityHealth` component have the statistics referenced in your game's `DamageSourceSOs`.

A possible way to simplify the management of all the `DamageSourceSO` modifier stats is to create a `StatSet` specifically for this purpose, and include it as a *Included Stat Set* in the various entities' `StatSets` of your game. This way, you centralize all the `DamageSourceSO` modifier stats in a single `StatSet`, and you can easily keep track of them and ensure that they are included in all the relevant entities.

Damage Types

A **DamageTypeSO** represents the type of damage—such as physical, fire, ice, lightning, or damage-over-time (DoT) effects like bleeding. In the following image you can see an example of a **DamageTypeSO** instance in the inspector:



You can notice that the parameters are divided in three sections:

1. **Damage Reduction:** parameters related to the damage and defense reduction mechanics for this damage type.
2. **Damage Modifiers:** parameters related to flat and percentage damage modifiers for this damage type.
3. **True Damage Options:** parameters related to the true damage options for this damage type.

NOTE

I recall the [Damage Modifiers vs. Stat-Based Damage Reduction](#) section of the introduction documentation, where I explained the difference between damage reduction and damage modifiers.

We will now see each of these sections in detail.

Damage Type's Damage Reduction

The primary use case for **DamageTypeSO** is implementing entities with varying resistances to specific damage types. This is primarily achieved via **Defensive Stats**. For each **DamageTypeSO**, you can define a defensive statistic that reduces incoming damage of that type. For example, an **Armor** stat might reduce **Physical** damage, while a **Magic Resistance** stat reduces **Magic** damage.

The value of the defensive stat is fed into the associated **Damage Reduction Fn** (function) and used to calculate the actual damage reduction. The package provides some built-in damage reduction functions, such as:

- **Flat Dmg Reduction:** Reduces damage by a flat amount equal to the defensive stat value multiplied by a constant.
- **Percent Dmg Reduction:** Reduces damage by a percentage equal to the defensive stat value.
- **Log Dmg Reduction:** Reduces damage in a logarithmic way based on the defensive stat value, providing diminishing returns as the stat increases.

NOTE

Defensive Stat and **Damage Reduction Fn** are optional. However, they must always be configured together: if one is set, the other must be set as well. If only one of them is assigned, a warning will be logged at runtime and the damage reduction step will be skipped for that **DamageTypeSO**.

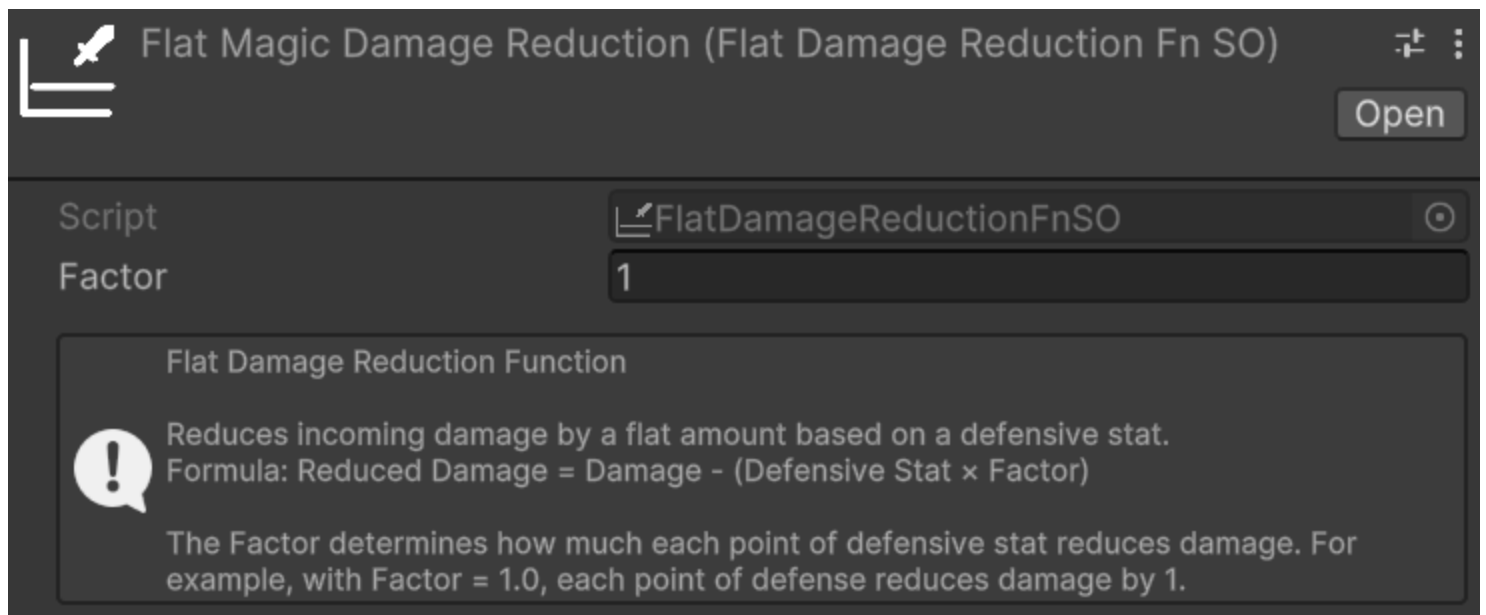
WARNING

If the target entity lacks the statistic referenced by **Defensive Stat**, an error will be logged when applying damage of that type. Ensure that all entities with an **EntityHealth** component have the defensive statistic referenced by the **DamageTypeSOs** used in your game.

Let's see all of them in detail.

Damage Reduction Functions - Flat Dmg Reduction

Relative path: **Dmg Reduction Functions** -> **Flat Dmg Reduction**



The flat damage reduction function is the most simple and straightforward one. It reduces damage by a flat amount equal to the defensive stat value multiplied by the **Factor** specified via the Inspector.

For example:

- **Damage Type:** Magic damage.
- **Defensive Stat for the Magic Damage Type:** Magic Resistance.
- **Damage Reduction Function for the Magic Damage Type:** Flat Dmg Reduction with a scaling factor of 2.

In this case, if an entity has Magic Resistance equal to 10, and is about to take 50 Magic damage, the damage reduction will be equal to 10 (Magic Resistance) * 2 (scaling factor) = 20. So the final damage taken by the entity will be 50 (initial damage) - 20 (damage reduction) = 30 Magic damage. Clearly, this example assumes that there are no other damage modifications (e.g., neutral damage modifiers).

Use Cases: This function is best suited for games where offensive and defensive stat values are low — close to single digits or tens at most. In these scenarios, the linear relationship between the defensive stat and the damage reduction makes it trivial to ensure that no entity can completely negate incoming damage, as long as the stat values remain within a controlled range. Percentage or logarithmic reduction functions may yield imprecise or unintuitive results at such coarse-grained stat scales.

Pros:

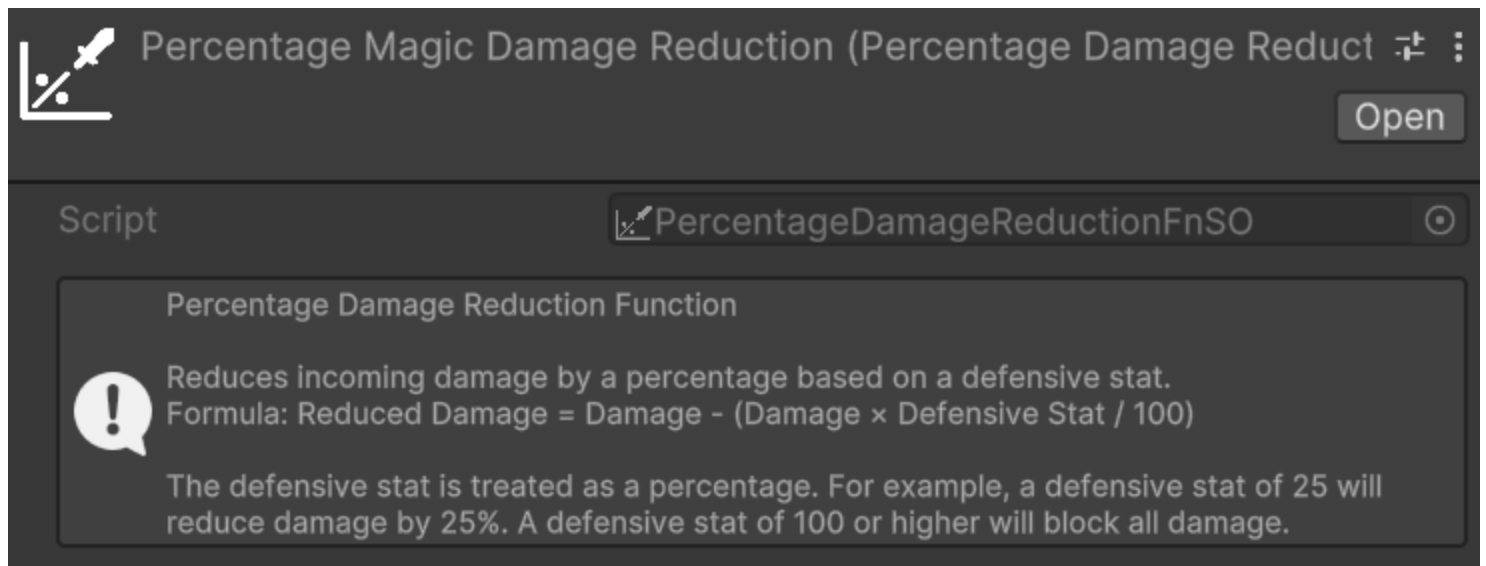
- Damage reduction is highly predictable: knowing the defensive stat value and the factor, anyone can instantly calculate the resulting reduction.
- Simple to debug: the math is entirely transparent, making it straightforward to verify that the damage pipeline is working as expected at every step.
- Easy to balance: the linear relationship between the stat and the reduction makes it trivial to tune the factor to achieve the desired game feel.

Cons:

- Simplistic system: this function may not suit games that require nuanced or complex defensive mechanics.
- Risk of complete damage negation: if defensive stat values grow too high relative to typical damage values, entities can become entirely immune to certain damage types. This can happen, for example, if the level difference between attacker and defender is too high.

Damage Reduction Functions - Percent Dmg Reduction

Relative path: `Dmg Reduction Functions -> Percentage Dmg Reduction`



The percentage damage reduction function reduces incoming damage by a percentage equal to the defensive stat value. For example, if an entity has a defensive stat of 30, it will receive 30% less damage of the associated type.

Use Cases:

- Games where players can face enemies with notable level or power differences. Because the reduction is percentage-based, even a low-level entity with a small but non-zero defensive stat will always receive a proportional reduction, preventing extreme damage scenarios that would arise from large stat disparities between the attacker and the defender.
- Systems that need to be less sensitive than the Flat Dmg Reduction to the exact magnitude of offensive and defensive stats, while still remaining predictable and easy to balance.

Pros:

- Predictable and straightforward to reason about: a stat value of X directly translates to X% less damage.
- Remains effective regardless of the magnitude of incoming damage: even against a significantly stronger attacker, the same proportional reduction applies, guaranteeing that defense always has a

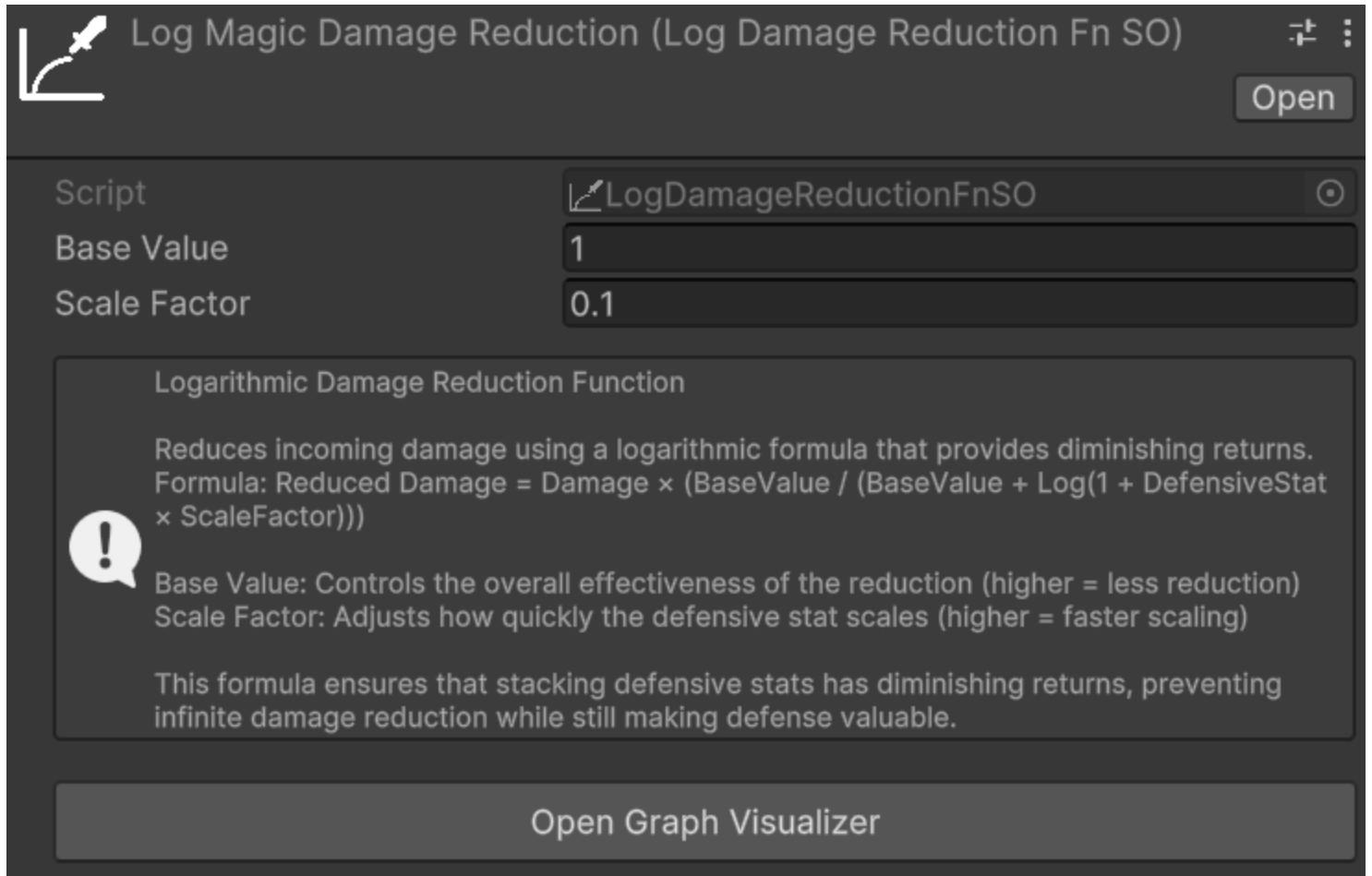
meaningful impact.

Cons:

- High risk of damage immunity at extreme values: once the defensive stat reaches 100, the entity becomes completely immune to that damage type. This can be especially problematic for tank-oriented entities or builds designed to stack defensive stats.
- Can make late-game balancing challenging if stat values are not tightly bounded.

Damage Reduction Functions - Log Dmg Reduction

Relative path: Dmg Reduction Functions -> Log Dmg Reduction



Log Magic Damage Reduction (Log Damage Reduction Fn SO) Open

Script LogDamageReductionFnSO

Base Value 1

Scale Factor 0.1

Logarithmic Damage Reduction Function

Reduces incoming damage using a logarithmic formula that provides diminishing returns. Formula: $\text{Reduced Damage} = \text{Damage} \times (\text{BaseValue} / (\text{BaseValue} + \log(1 + \text{DefensiveStat} \times \text{ScaleFactor})))$

! Base Value: Controls the overall effectiveness of the reduction (higher = less reduction)
Scale Factor: Adjusts how quickly the defensive stat scales (higher = faster scaling)

This formula ensures that stacking defensive stats has diminishing returns, preventing infinite damage reduction while still making defense valuable.

Open Graph Visualizer

The logarithmic damage reduction function reduces damage using a logarithmic curve, providing diminishing returns as the defensive stat value increases. A small initial investment in the defensive stat yields a substantial reduction, while further investment produces progressively smaller gains. This makes it theoretically impossible to reach 100% reduction regardless of how high the stat grows.

The Formula

The logarithmic damage reduction function applies the following formula to compute the final damage taken:

Reduced **Damage** = **Damage** × (BaseValue / (BaseValue + **Log**(1 + DefensiveStat × ScaleFactor)))

Two parameters, configurable directly on the **Log Dmg Reduction** asset, control the shape of the curve:

- **Base Value:** the constant in the denominator of the reduction multiplier. A larger Base Value reduces the relative weight of the logarithmic term, making the reduction curve less aggressive — the same defensive stat value will produce less damage reduction. Conversely, a smaller Base Value amplifies the effect of the logarithm, yielding stronger reductions even at lower stat values.
- **Scale Factor:** the multiplier applied to the defensive stat value before computing the logarithm. A larger Scale Factor causes the curve to climb more steeply at low stat values, reaching substantial reductions earlier. A smaller Scale Factor stretches the curve, requiring higher stat values to achieve the same reduction.

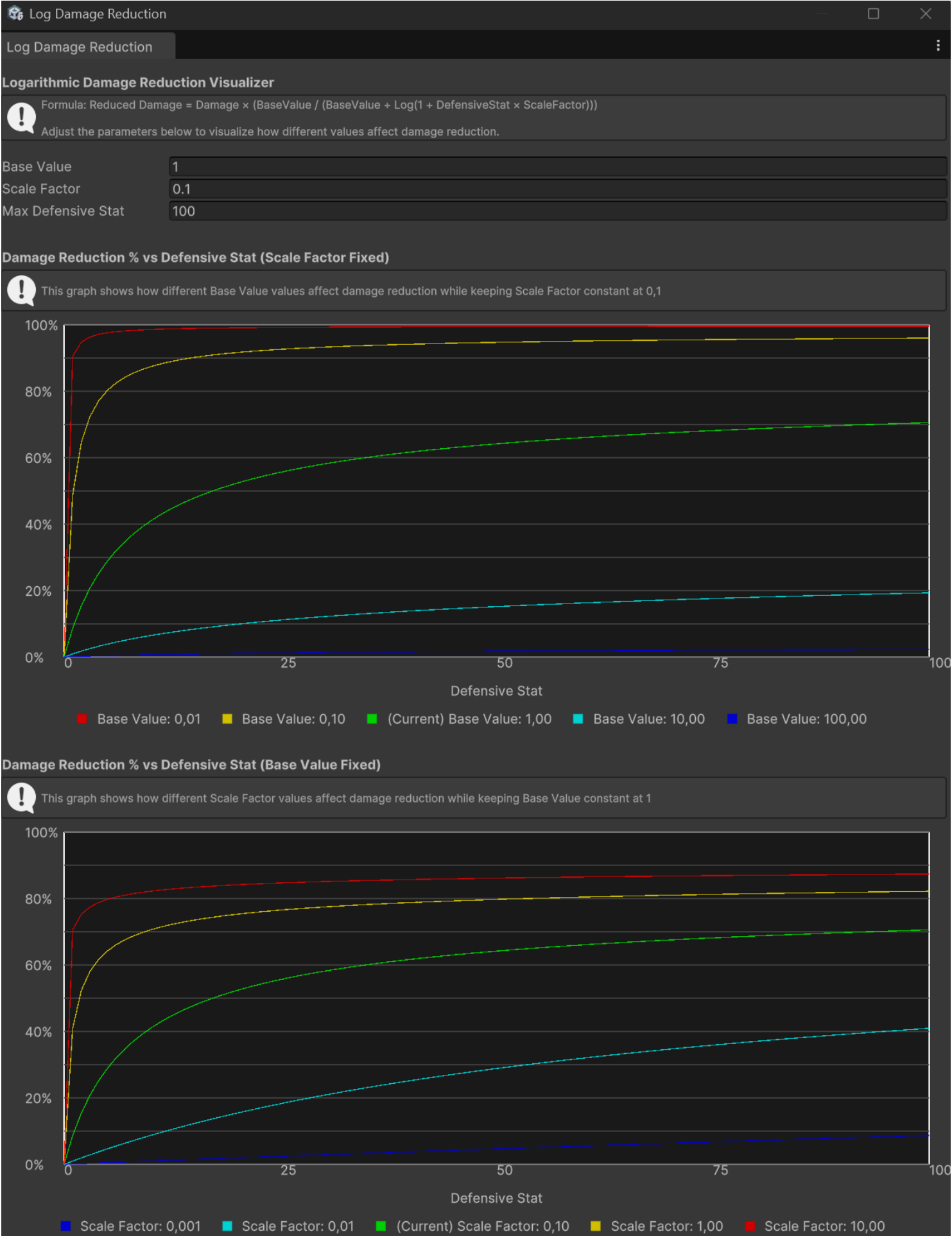
Both parameters must be set to strictly positive values.

Log Damage Reduction Graph

Because the non-linear nature of the formula makes it difficult to reason about the effect of Base Value and Scale Factor at a glance, the package includes a dedicated visualization window. You can open it in two ways:

- Clicking the **Open Graph Visualizer** button in the inspector of any **Log Dmg Reduction** asset.
- From the Unity menu: **Window → Astra RPG Health → Log Damage Reduction Graph**.

The graph visualizer window should look like this:



Once the window is open, three fields let you configure the visualization:

- **Base Value:** the reference Base Value to analyze, matching the value you have set on your asset.
- **Scale Factor:** the reference Scale Factor to analyze, matching the value you have set on your asset.
- **Max Defensive Stat:** the upper bound of the X-axis, i.e., the maximum defensive stat value to plot.

The window displays two separate graphs, each plotting **Damage Reduction %** on the Y-axis and the **Defensive Stat value** on the X-axis:

1. **Scale Factor fixed, Base Value varying** — shows five curves corresponding to different Base Values centered around the reference value you configured, while Scale Factor is held constant. This lets you compare how increasing or decreasing Base Value shifts the reduction curve, making it easier to find a value that achieves the desired behavior for your game's stat range.
2. **Base Value fixed, Scale Factor varying** — shows five curves corresponding to different Scale Factors centered around the reference value, while Base Value is held constant. This lets you compare how Scale Factor affects the steepness of the initial climb of the curve.

In both graphs the legend labels each curve with its exact parameter value, and the middle curve (marked *Current*) corresponds to the reference value you entered. Hovering the mouse over either graph shows a tooltip with the exact damage reduction percentage produced by each curve for the defensive stat value under the cursor, like this:



Use Cases:

- RPGs with wide stat ranges and long progression curves — for example, games with levels 1 through 100 or beyond — where both offensive and defensive stats grow substantially over time. The diminishing returns ensure that no entity can become completely immune to a damage type simply by stacking the defensive stat.
- Games where investing in defense should always be viable, but never dominant: players are rewarded for defensive investment, yet the diminishing returns naturally discourage over-specialization and keep combat meaningful at all stages.
- Projects that need a self-capping damage reduction formula without enforcing a hard maximum: the logarithmic curve naturally prevents extreme values from causing damage immunity, reducing

the need for manual clamping or caps in the game design.

Pros:

- Inherently prevents complete damage immunity: the logarithmic curve approaches but never actually reaches 100% reduction, no matter how high the defensive stat grows.
- Scales gracefully across wide stat ranges, remaining meaningful at every stage of the game without requiring constant rebalancing.
- Discourages over-specialization in defense: the diminishing returns act as a natural soft cap, making it progressively less efficient to stack defensive stats beyond a certain point.

Cons:

- Less intuitive than flat or percentage reduction: players and designers cannot easily predict the exact damage reduction for a given stat value at a glance — the log graph window tool is typically needed.
- More complex to debug and tune: the non-linear relationship between the stat and the reduction requires more careful analysis and testing during development.

Damage Reduction Functions - Custom Dmg Reduction Functions

If you want to provide your own custom damage reduction function, you can create a new class that inherits from [DamageReductionFnSO](#) and implement the `CalculateReducedDamage` method. Remember to use the `CreateAssetMenu` attribute (or the `MenuItem` attribute) to make it creatable from the Unity editor. You can take a look at the existing damage reduction functions implementations for reference.

Defense Penetration

Defense penetration allows the damage dealer to partially bypass the target's defensive stat before damage reduction is calculated. This mechanism is useful for implementing mechanics such as armor penetration or magic penetration, where an attacker can reduce the effective defenses of the target.

Two optional parameters of the `DamageTypeSO` control this behavior:

- **Defensive Stat Pierced By:** the statistic on the **damage dealer** that pierces the target's defensive stat. For example, an `Armor Penetration` stat might pierce the `Armor` stat of the target.
- **Defense Reduction Fn:** the function that computes how the piercing stat lowers the target's defensive stat. The resulting reduced defensive stat value is then passed to the **Damage Reduction Fn** in place of the original value.

NOTE

Defensive Stat Pierced By and **Defense Reduction Fn** are optional. However, they must always be configured together: if one is set, the other must be set as well. If only one of them is assigned, a warning will be logged at runtime and the defense penetration step will be skipped for that `DamageTypeSO`.

WARNING

If the damage dealer entity lacks the statistic referenced by **Defensive Stat Pierced By**, an error will be logged when applying damage of that type. Ensure that all entities capable of dealing damage of this type have the relevant piercing statistic.

To illustrate how defense penetration interacts with the rest of the damage pipeline, consider the following example:

- **Damage Type:** Physical damage.
- **Defensive Stat for the Physical Damage Type:** Armor.
- **Damage Reduction Function for the Physical Damage Type:** Flat Dmg Reduction with a factor of 1.
- **Defensive Stat Pierced By:** Armor Penetration.
- **Defense Reduction Fn:** Flat Def Reduction with a factor of 1.

In this case, if the target has Armor equal to 30 and the attacker has Armor Penetration equal to 10, the effective Armor value fed into the Damage Reduction Fn will be $30 (\text{Armor}) - 10 (\text{Armor Penetration}) \times 1 (\text{factor}) = 20$. So, if the incoming damage is 80, the final damage taken will be $80 - 20 (\text{effective Armor}) = 60$ Physical damage. This example assumes no other damage modifications are active.

The package provides three built-in Defense Reduction Functions — Flat, Percentage, and Logarithmic — that work in a fully analogous way to their counterparts described in the [Damage Reduction](#) section above. The parameters, trade-offs, and use cases of each variant are the same; the only difference is that these functions operate on the **defensive stat value** rather than directly on the damage amount. For a detailed description of each, refer to the [Damage Reduction](#) section.

Relative path: Def Reduction Functions -> Flat Def Reduction

 Flat Def Reduction Fn (Flat Defense Reduction Fn)   

Open

Script  FlatDefenseReductionFn 

Factor 1

Flat Defense Reduction Function

 Reduces a target's defensive stat by a flat amount based on a piercing stat.
Formula: $\text{Reduced Defense} = \text{Defense} - (\text{Piercing Stat} \times \text{Factor})$

The Factor determines how much each point of piercing stat reduces defense. For example, with Factor = 1.0, each point of piercing reduces defense by 1.

Relative path: Def Reduction Functions -> Percentage Def Reduction

 Percentage Def Reduction Fn (Percentage Defense Reduction Fn)   

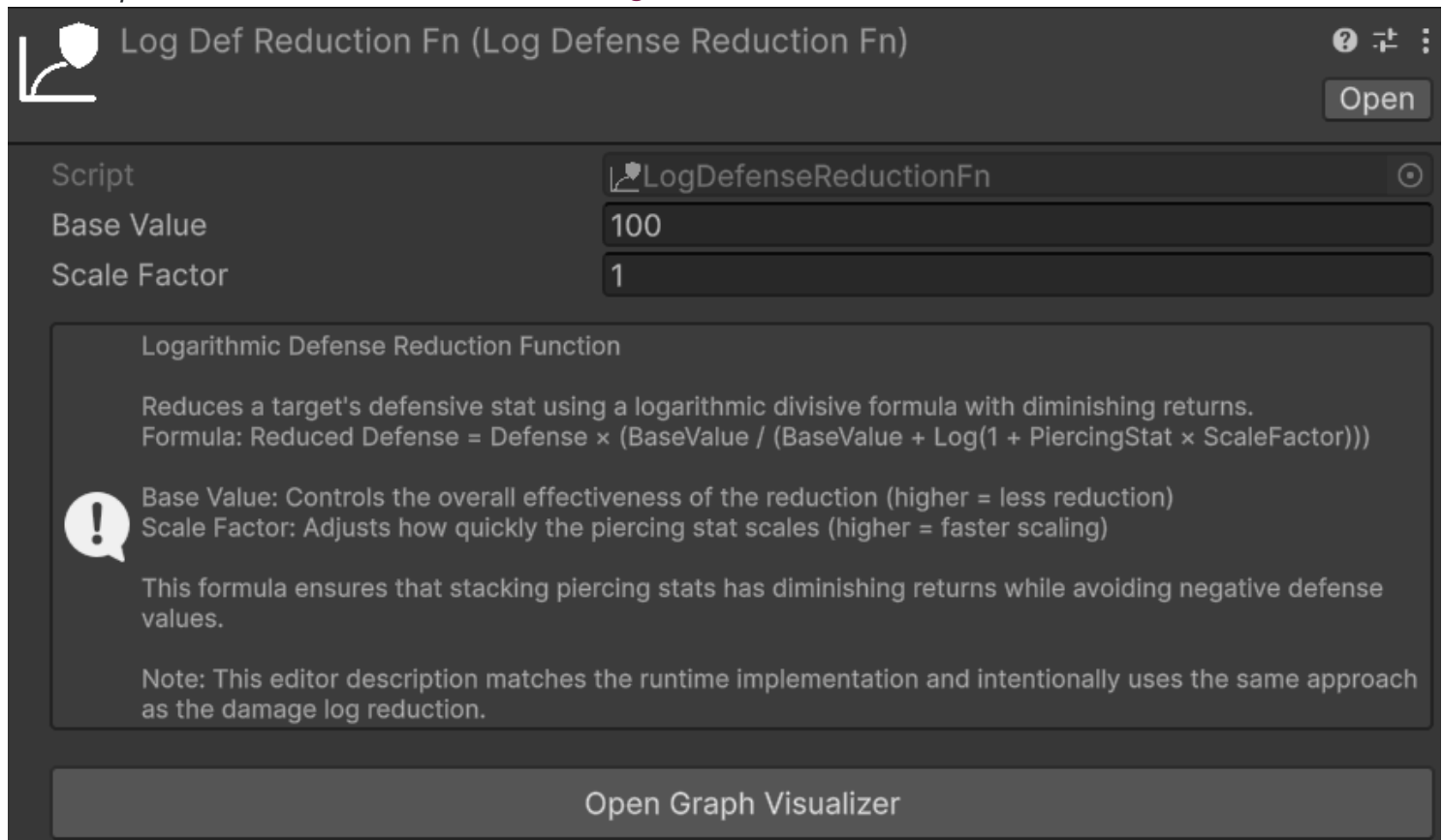
Open

Script  PercentageDefenseReductionFn 

Percentage Defense Reduction Function

 Reduces a target's defensive stat by a percentage based on a piercing stat.
Formula: $\text{Reduced Defense} = \text{Defense} - (\text{Defense} \times \text{Piercing Stat} / 100)$

The piercing stat is treated as a percentage. For example, a piercing stat of 30 will reduce defense by 30%. A piercing stat of 100 or higher will completely negate all defense.



If none of the built-in Defense Reduction Functions suits your needs, you can implement a custom one by creating a class that inherits from [DefenseReductionFnSO](#) and implementing the `CalculateReducedDefense` method. As with the Damage Reduction Functions, remember to use the `CreateAssetMenu` attribute (or the `MenuItem` attribute) to make it creatable from the Unity editor.

Damage Type's Damage Modifiers

The **Percentage Modifier Stat** and **Flat Modifier Stat** fields in this section let you assign stat-based damage modifiers that apply specifically when an entity receives damage of this `DamageTypeSO`. Assigning a positive value to either stat increases the damage received from this type; a negative value decreases it. For a full explanation of how all damage modifier categories work and how they stack together, see the [Damage Modifiers](#) section.

A possible way to simplify the management of all the `DamageTypeSO` modifier stats is to create a `StatSet` specifically for this purpose, and include it as a *Included Stat Set* in the various entities' `StatSets` of your game. This way, you centralize all the `DamageTypeSO` modifier stats in a single `StatSet`, and you can easily keep track of them and ensure that they are included in all the relevant entities.

Damage Type's True Damage Options

The **True Damage Options** section of a `DamageTypeSO` exposes three boolean flags that allow selective bypasses for individual stages of the damage pipeline. They are the primary tool for implementing "true

damage" mechanics — damage that partially or entirely skips specific mitigation layers — without requiring a custom `DamageCalculationStrategy`.

- **Ignore Barrier:** when enabled, the [ApplyBarrierStep](#) is skipped entirely for this `DamageTypeSO`. Damage bypasses the target's barrier and is applied directly to its health pool. Use this for damage types that are meant to be unavoidable through shielding — for example, a pure true damage type, fall damage, or damage-over-time effects that should not interact with temporary shields.
- **Ignore Generic Percentage Modifiers:** when enabled, the [ApplyPercentageDmgModifiersStep](#) skips the **Generic Percentage Damage Modification Stat** configured in `AstraRpgHealthConfigSO`. Percentage modifiers from the `DamageSourceSO` or from this `DamageTypeSO` itself still apply normally.
- **Ignore Generic Flat Modifiers:** when enabled, the [ApplyFlatDmgModifiersStep](#) skips the **Generic Flat Damage Modification Stat** configured in `AstraRpgHealthConfigSO`. Flat modifiers from the `DamageSourceSO` or from this `DamageTypeSO` itself still apply normally.

NOTE

Ignore Generic Percentage Modifiers and **Ignore Generic Flat Modifiers** bypass only the *generic* modifier layer — the global stats configured in `AstraRpgHealthConfigSO`. Source-specific and type-specific modifier stats are never bypassed by these flags and always participate in the pipeline normally.

However, if a `DamageSourceSO` or this `DamageTypeSO` has no modifier stats assigned, those layers contribute nothing by construction — which effectively produces the same result as a bypass. Enabling all three flags while also leaving all modifier stats unset on the source and type produces a fully modifier-free damage type.

TIP

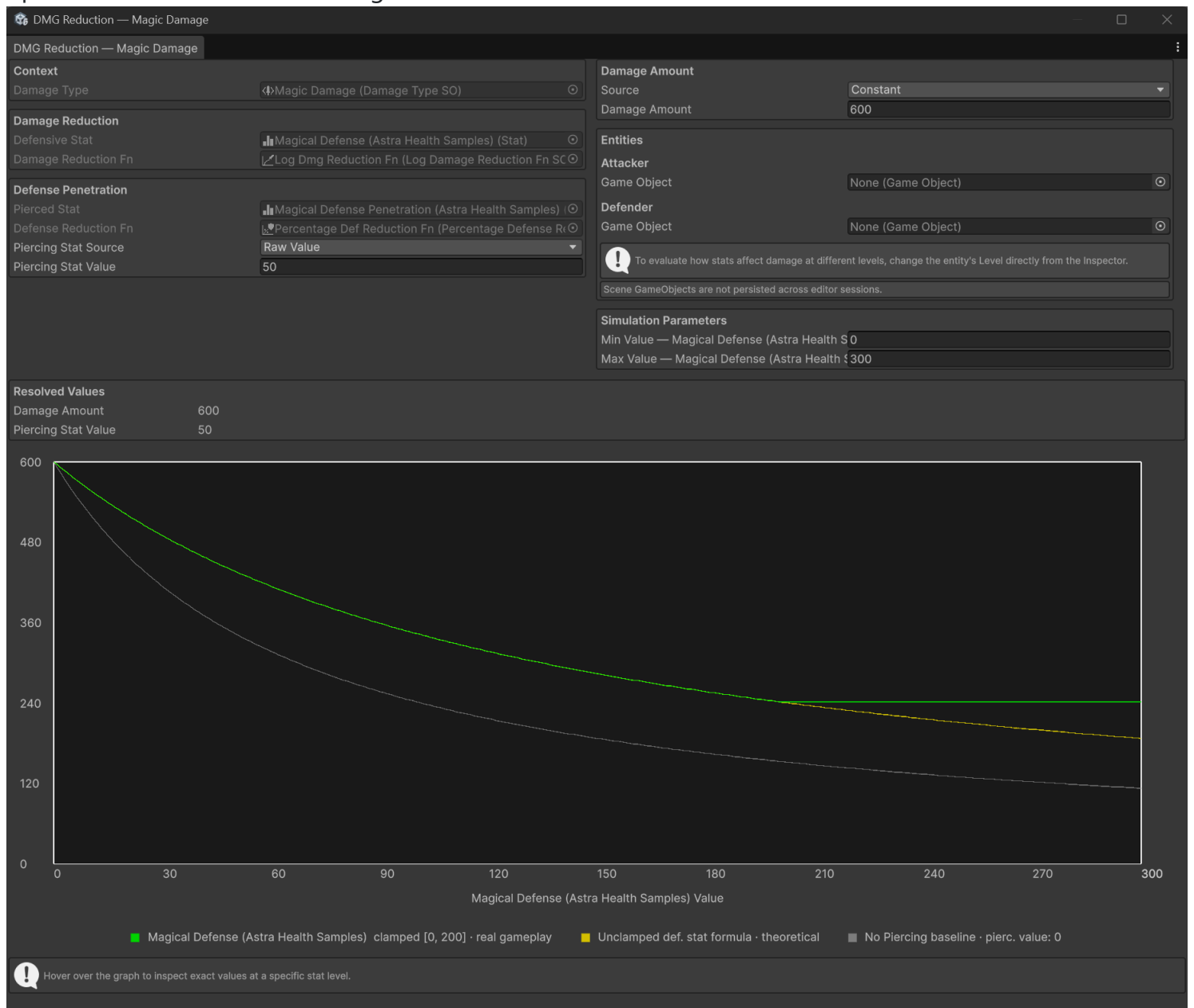
To create a damage type that is also unaffected by any defensive stat, simply leave **Defensive Stat** and **Damage Reduction Fn** empty. The [ApplyDefenseStep](#) has nothing to compute and is skipped. Combined with the three True Damage Option flags and no modifier stats, this defines a fully unmitigated damage type.

Damage Reduction Graph

The **Damage Reduction Graph** is an Editor window that plots how damage received scales as a function of a defender's defensive stat value. Use it to visually verify the behavior of your damage reduction and

defense penetration configuration, preview the impact of armor penetration at different stat values, and compare clamped (defensive stat) vs. unclamped (defensive stat) curves.

To open it, click the **Open Damage Reduction Graph** button in the inspector of any **DamageTypeSO**. Once opened, it should look something like this:



Obviously the plot and the values will vary based on the specific configuration of the **DamageTypeSO** you have.

All configuration fields — **Defensive Stat**, **Damage Reduction Fn**, **Pierced Stat**, and **Defense Reduction Fn** — are read-only inside the window and reflect the asset directly. To change any of them, edit the **DamageTypeSO**.

i NOTE

The Damage Reduction Graph is a **simulation tool only**. No asset or scene object is modified when you change values inside the window.

Control Panel

The window is divided into two equal columns that scale with the window width. Below them are the **Resolved Values** panel, an optional **Defender Breakdown** panel (shown only when a Defender is assigned in the Entities section), and the graph itself.

Context (read-only)

Displays the **DamageTypeSO** that opened the window as a disabled field. It cannot be reassigned here — to analyze a different asset, open its graph from its own inspector.

Damage Reduction

Field	Description
Defensive Stat	The stat whose value sweeps the X axis of the graph. Read-only — sourced directly from the DamageTypeSO . To change it, edit the asset.
Damage Reduction Fn	The DamageReductionFnSO that converts (damage , effectiveDefense) into finalDamage . Read-only — sourced directly from the DamageTypeSO . To change it, edit the asset.

If either field is left empty, a warning is shown and the graph is not drawn.

Defense Penetration

Field	Description
Pierced Stat	The stat on the attacker that provides armor penetration, read-only — sourced from DamageTypeSO.DefensiveStatPiercedBy . To change it, edit the asset.
Defense Reduction Fn	Optional DefenseReductionFnSO that reduces the defender's stat before damage reduction is applied. Read-only — sourced directly from the DamageTypeSO . To change it, edit the asset. When assigned, a Piercing Stat Source selector appears.

The **Piercing Stat Source** controls where the piercing stat value comes from:

Source	Description
Raw Value	A plain number field. Type the piercing stat value directly.
Growth Formula	A GrowthFormula asset plus a Level field clamped to the formula's valid range. The resolved value is displayed inline.
Class at Level	A Class asset with either a constant level integer or a runtime IntVar . The piercing stat is read from DamageTypeSO.DefensiveStatPiercedBy for that class at the given level. Requires Defensive Stat Pierced By to be set on both the Class and DamageTypeSO assets.
From Attacker	Reads DamageTypeSO.DefensiveStatPiercedBy directly from the Attacker entity set in the Entities section. Requires Defensive Stat Pierced By to be set on the DamageTypeSO asset and an Attacker to be assigned. The Attacker must also have the Defensive Stat Pierced By stat in its EntityStats component.

NOTE

If no **Defense Reduction Fn** is assigned, the Piercing Stat Source controls are hidden and the grey baseline line is not drawn on the graph.

Damage Amount

Source	Description
Constant	A plain number field. Defaults to 100.
Scaling Formula	A ScalingFormula asset evaluated against the Attacker and/or Defender entities configured in the Entities section. Info panels in the window list which entity components the formula requires.

Entities

Field	Description
Attacker	An in-scene GameObject with EntityCore . Required when Damage Amount source is Scaling Formula or Piercing Stat Source is From Attacker .
Defender	An in-scene GameObject with EntityCore and EntityStats . When assigned, the Defender Breakdown panel appears below the Resolved Values panel.

TIP

To simulate how the graph changes at different entity levels, modify the entity's **Level** field directly in the Inspector while the window is open. The graph updates in real time. Level controls are intentionally absent from the window itself to avoid triggering **OnEntityLevelUp** / **OnEntityLevelDown** events while interacting with the graph window.

Simulation Parameters

Field	Description
Min Value — {stat name}	Left bound of the X axis. Defaults to 0. Always editable, regardless of whether the Defensive Stat has a Min Value defined.
Max Value — {stat name}	Right bound of the X axis. Always editable.

Reading the Graph

The graph plots **Damage Received** on the Y axis and the **Defensive Stat Value** on the X axis. Up to three lines are drawn simultaneously:

Line	Color	Condition	What it represents
Real gameplay	Green	Always	Damage received as the stat grows, with the stat clamped to its configured bounds (if any). This is the primary reference curve — it reflects the exact behavior any entity can exhibit in-game. When the stat has no bounds configured, this line coincides with the unclamped formula and is the only curve shown.

Line	Color	Condition	What it represents
Unclamped formula	Yellow	Only when the Defensive Stat has a Min Value and/or Max Value	Same formula but with the stat not clamped to its bounds — useful for understanding the theoretical shape of the reduction function and how much the configured bounds constrain it. Only shown when it meaningfully differs from the green line. It may happen that you still see only the green line if the bounds do not affect the outcome (aka. the two lines coincide).
No-piercing baseline	Grey	Only when Defense Reduction Fn is assigned and effective piercing $\neq 0$	The damage curve with piercing forced to zero. Lets you immediately see how much the configured penetration shifts the outcome relative to a no-penetration scenario.

The legend below the graph labels each active line with its color and — for the green line — the exact bound(s) at which the stat is clamped.

Hover tooltip — moving the mouse over the graph shows a tooltip with, for each active line, the damage received and the absolute and percentage reduction at the hovered stat value. The detail rows also show the raw defensive stat value at the cursor and, when a **DefenseReductionFnS0** is set, the portion of the stat ignored by piercing and the resulting effective defense.

It should look like this:



Resolved Values and Defender Breakdown

The **Resolved Values** panel, always visible below the control columns, shows the values actually fed into the graph — the resolved damage amount and the resolved piercing stat value. If either resolution fails (e.g., a required entity is missing or a formula throws an exception), an error message is shown here and the graph is suppressed.

When a **Defender** is assigned with **EntityCore** and **EntityStats**, the **Defender Breakdown** panel appears alongside the Resolved Values panel. It provides a point-in-time snapshot of the exact damage calculation for the defender at its current in-scene stat values:

Row	Description
Damage Amount	The resolved incoming damage
Piercing Stat Value	The resolved piercing value
{stat.name} (defender)	The defender's current defensive stat value
Defense Reduced by Piercing	How much of the defensive stat is bypassed by the piercing value (only when Defense Reduction Fn is set)
Effective {stat.name}	The final defensive stat fed into the damage reduction function (only when Defense Reduction Fn is set)
Final Damage Received	The final damage the defender would receive at its current level and stats
Damage Reduction (abs / %)	The absolute and percentage difference between raw and final damage

The breakdown reflects the Defender's **current** in-scene stat values at its current level. To see how the numbers change at a different level, modify the entity's Level field directly in the Inspector as described above.

Here is a screenshot of the Defender Breakdown panel in action:

Damage Amount

Source

Scaling Formula (based on Entities Section)

Damage Formula

↗

Slice and Dice Scaling Fo (Scaling Formula)

Attacker (self) needs: StatsScalingComponent

Uses Attacker (self) and Defender (target) from the Entities section.

Entities

Attacker

Game Object

Assassin

Defender

Game Object

Dummy (with EntityHealth)

!

To evaluate how stats affect damage at different levels, change the entity's Level directly from the Inspector.

Scene GameObjects are not persisted across editor sessions.

Simulation Parameters

Min Value — Armor (Astra Health Samples)

0

Max Value — Armor (Astra Health Samples)

300

Defender — Dummy (with EntityHealth)

Damage Amount

385

Piercing Stat Value

50

Armor (Astra Health Samples) (defender)

150

Defense Reduced by Piercing

75

Effective Armor (Astra Health Samples)

75

Final Damage Received

182,00

Damage Reduction (abs)

203,00

Damage Reduction (%)

52,7%

Notice that a `ScalingFormula` is used for the damage amount, the *Dummy* entity set as Defender in the Entities section, and the Defender Breakdown panel on the bottom showing the exact damage calculation for the Defender based on its current stats and level.

Dealing Damage to an Entity

The API method you will use the most with this package is certainly `TakeDamage`, whose responsibility is to apply damage to the entity, taking into account modifiers, immunity, the damage calculation strategy, and other relevant mechanics. This method takes a `PreDamageContext` as input.

The recommended way via code to inflict damage on an entity is as follows:

1. Construct an instance of `PreDamageContext` with all relevant information about the damage you intend to inflict through its fluent builder.
2. Call `TakeDamage` passing the newly constructed context.

Suppose we are implementing a skill that deals 50 fire damage to the target. The code to apply damage to the target could be the following:

```
// Assuming that:
// - dmgType is a DamageType representing fire damage
// - dmgSource is a DamageSource representing the damage coming from a skill
// - target is the EntityCore that we want to damage
// - skillCaster is the EntityCore that casts the skill

// first we ensure that the target has an EntityHealth component
if (target.TryGetComponent(out EntityHealth targetHealth)) {
    // then we build the PreDamageContext with all the relevant information
    var preDamageContext = PreDamageContext.Builder
        .WithAmount(50)
        .WithType(dmgType)
        .WithSource(dmgSource)
        .WithTarget(target)
        .WithDealer(skillCaster)
        .Build();

    // finally, we call TakeDamage to apply the damage to the target
    targetHealth.TakeDamage(preDamageContext);
}
```

Thanks to the `PreDamageContext` fluent builder, the IDE will automatically suggest the fields to fill in one at a time. As long as it presents them one at a time, it means they are required fields. If instead it presents more than one at a time, it means those fields are optional, and you can decide whether to fill them in or build the context without them. Optional fields are, for example, the critical hit flag and the critical multiplier. In the example, for simplicity, I did not fill in these fields.

Now, we know that hardcoding the damage value directly in the code is not a good practice. Let's see how to use a `ScalingFormula` to dynamically calculate the amount of damage to inflict. The step builder creation would become the following:

```
// Assuming that scalingFormula is a ScalingFormula that calculates the damage amount based
// on the skill caster's stats...

// ...we calculate the damage amount by evaluating the scaling formula
long damageAmount = scalingFormula.CalculateValue(skillCaster);
var preDamageContext = PreDamageContext.Builder
```

```
.WithAmount(damageAmount)
.WithType(dmgType)
.WithSource(dmgSource)
.WithTarget(target)
.WithDealer(skillCaster)
.Build();
```

Damage Modifiers

Damage modifiers are a flexible tool for implementing mechanics such as resistances, weaknesses, buffs, and debuffs that affect how much damage an entity receives. Unlike [Defensive Stat-Based Damage Reduction](#), damage modifiers are off by default — their stats default to 0 — and can both increase and decrease the damage amount.

Three categories of damage modifiers exist, and they all stack additively with one another:

- **Generic modifiers:** apply to all damage received by an entity, regardless of damage type or source.
- **DamageSource modifiers:** apply only when the damage originates from a specific [DamageSourceSO](#).
- **DamageType modifiers:** apply only when the damage is of a specific [DamageTypeSO](#).

Generic Damage Modifiers

Generic modifiers are configured in the [AstraRpgHealthConfigSO](#) asset and apply universally to every instance of damage received by an entity. The two relevant fields are **Generic Flat Damage Modification Stat** and **Generic Percentage Damage Modification Stat**, described in detail in the [Package Configuration](#) page.

As a quick recap:

- **Generic Flat Damage Modification Stat:** a [Stat](#) whose value is added to (or subtracted from) the incoming damage amount as a flat quantity. A positive value increases the damage received; a negative value decreases it.
- **Generic Percentage Damage Modification Stat:** a [Stat](#) whose value is applied as a percentage modification to the incoming damage. A value of 20 means +20% more damage received; a value of -20 means -20% less damage received.

DamageSource Modifiers

As already introduced in the [Damage Sources](#) section, each [DamageSourceSO](#) asset exposes a **Percentage Modifier Stat** and a **Flat Modifier Stat**. These work identically to the generic modifiers described above, but are applied only when the damage originates from that specific [DamageSourceSO](#).

DamageType Modifiers

Similarly, each `DamageTypeSO` asset exposes a **Percentage Modifier Stat** and a **Flat Modifier Stat** in its **Damage Modifiers** inspector section. These work in the same way as the source-specific modifiers, but are applied only when the damage is of that specific `DamageTypeSO`. A typical use case is implementing elemental weaknesses and resistances: for example, a **Fire Weakness** stat could be assigned as the **Percentage Modifier Stat** of a Fire `DamageTypeSO`, so that entities with a positive value of that stat take proportionally more Fire damage.

⚠ WARNING

As with `DamageSourceSO` modifiers, if the target entity lacks any statistic referenced by the **Percentage Modifier Stat** or **Flat Modifier Stat** fields on a `DamageTypeSO`, an error will be logged when applying damage of that type. Ensure that all entities with an `EntityHealth` component have the damage modifier statistics referenced by the `DamageTypeSOs` used in your game. A practical approach is the same suggested for `DamageSourceSO` modifiers: centralize all modifier stats in a dedicated `StatSet` and include it in the relevant entities' stat sets.

Stacking Behavior

When the `ApplyFlatDmgModifiersStep` and `ApplyPercentageDmgModifiersStep` steps are included in the active `DamageCalculationStrategy` (that we will see soon in the [Damage Calculation Strategy](#) section), all applicable modifiers of the same kind are **summed additively** into a single net value, which is then applied to the current damage amount in one operation.

For percentage modifiers, the system also performs individual immunity checks before summing: if the generic percentage modifier alone reaches -100 or below, the damage is fully prevented and the entity is flagged as immune to all damage for that hit; if the source- or type-specific modifier alone reaches -100 or below, the damage is prevented and the entity is flagged as immune to that specific source or type respectively.

Damage Calculation Pipeline

When you call `TakeDamage` on an `EntityHealth`, the raw damage amount you specify does not reach the entity's health pool directly. Instead, it passes through the **Damage Calculation Pipeline**: an ordered sequence of processing steps, each responsible for one aspect of the calculation — barriers, critical multipliers, modifiers, and defenses. Think of it as a production line: raw damage enters at the first station and is transformed by each step in turn; what exits the last step is what the entity actually loses from its health.

The strength of this design is its configurability. Which steps are active, and in what order, is defined entirely by a `DamageCalculationStrategy` asset that you set up in the Inspector — no scripting required. Different entities can use different strategies: a boss might have its own pipeline featuring an extra final

step that limits the incoming damage to a maximum value of 10% of the Max HP, while regular enemies fall back to the project-wide default. Strategies can also be swapped at runtime, enabling mechanics such as temporary invincibility phases or damage rules that change mid-encounter.

Internally, the pipeline runs its steps sequentially on a shared state object and short-circuits the moment damage is marked as prevented: if a barrier fully absorbs the hit, or a modifier grants immunity, all remaining steps are skipped and the result is returned immediately.

Pipeline Data Types

The pipeline uses a small set of dedicated types to keep concerns cleanly separated — there is one type for what you request, one for what flows through the steps, and one for what you receive back. As a designer integrating `TakeDamage` into your game code, you will interact with two of these directly:

`PreDamageContext` to describe the damage you want to apply, and `DamageResolutionContext` to inspect the outcome. The others are managed internally by the pipeline and are primarily relevant if you are implementing custom steps or hooking into damage events for analytics or reactions.

`PreDamageContext` is the descriptor you build before calling `TakeDamage`. Its mandatory fields are enforced by a fluent step-builder that requires them in order: `amount` → `type` → `damage source` → `target` → `source/dealer`. Optional fields include `IsCritical`, `CriticalMultiplier`, and `Ignore`. Setting `Ignore = true` causes the pipeline to be bypassed entirely; the damage is flagged as prevented with `PrePhaseIgnored` before any calculation begins.

`DamageInfo` is the mutable state object that flows through the pipeline. It is constructed from a `PreDamageContext` at the start of `TakeDamage` and holds:

- **Amounts** — a `DamageAmountContext` tracking the current damage value and step-by-step history.
- **Type, DamageSourceSO, Target, Source, IsCritical, CriticalMultiplier** — metadata forwarded from the `PreDamageContext`.
- **Reasons** — a `DamagePreventionReason` flags enum accumulating all reasons why damage was prevented.
- **IsPrevented** — returns `true` when `Reasons` is not `None`; used to short-circuit the pipeline.

`DamageAmountContext` tracks the numerical amount throughout the pipeline:

- **InitialAmount** — the original raw value; never modified after construction.
- **Current** — the damage value as modified by each step; steps read and write this property.
- **Records** — a read-only list of `StepAmountRecord` entries, each capturing the step type and the before/after `Current` values for that step. Useful for debugging and diagnostics.

`DamageResolutionContext` is the value returned by `TakeDamage`. It contains:

- **Outcome** — `DamageOutcome.Applied` or `DamageOutcome.Prevented`.
- **Reasons** — the accumulated `DamagePreventionReason` flags when prevented; `None` when applied.

- **TerminationStepType** — the **Type** of the step that caused early termination, if any.
- **FinalDamageInfo** — the **DamageInfo** at the end of the pipeline; may be **null** when damage was prevented in the pre-phase.
- **PreDamageContext** — the original input that initiated this damage attempt.

DamageOutcome is a two-value enum: **Applied** (damage was applied to the entity's health or barrier) and **Prevented** (damage was stopped before affecting the entity).

DamagePreventionReason is a flags enum that accumulates one or more reasons why damage was prevented:

Value	When set
None	No prevention; damage can proceed.
EntityImmune	Target has global immunity (IsImmune = true on EntityHealth).
AllDamageImmune	Generic percentage damage reduction stat alone reached $\leq -100\%$.
DamageTypeImmune	Type-specific percentage modifier alone reached $\leq -100\%$.
DamageSourceImmune	Source-specific percentage modifier alone reached $\leq -100\%$.
BarrierAbsorbed	Barrier fully absorbed the incoming damage.
DefenseAbsorbed	Defensive stat computation reduced damage to zero.
PrePhaseIgnored	PreDamageContext.Ignore was true before the pipeline started.
PrePhaseZeroAmount	PreDamageContext.Amount was zero or negative.
PipelineReducedToZero	A step reduced damage to zero without a more specific absorption reason.
EntityDead	Target was already dead when TakeDamage was called.

NOTE

DamagePreventionReason is a flags enum: multiple values can be set simultaneously on the same damage attempt. Inspect **DamageResolutionContext.Reasons** to read all accumulated reasons after a call to **TakeDamage**.

Damage Step

Each station in the pipeline is a `DamageStep` — an independent, composable unit of work that applies one focused transformation to the damage in flight. The five built-in steps cover the most common scenarios out of the box. If your game needs specialised behaviour — for example, a step that clamps the damage to a maximum value of 10% of the Max HP — you can implement a custom step by creating a class that inherits from `DamageStep` and implements `ProcessStep()`.

Step order matters. Each step receives the output of the one before it, so a different arrangement of the same steps produces a different result. Placing a critical multiplier step before defensive steps amplifies the pre-mitigation damage; placing it after scales a lower, post-mitigation value. We will see how to compose and reorder steps from the Inspector in the [Damage Calculation Strategy](#) section.

`DamageStep` is the abstract base class for all pipeline steps. Each step performs one focused transformation on `DamageInfo`. The pipeline runner calls `Process()` on each step in order; concrete implementations override `ProcessStep()`.

`Process()` enforces two preconditions automatically before delegating to `ProcessStep()`:

- If `DamageInfo.IsPrevented` is `true`, the step is skipped.
- If `DamageInfo.Amounts.Current` is ≤ 0 , the step is skipped.

After `ProcessStep()` returns, `Process()` appends a `StepAmountRecord` to `DamageAmountContext.Records` with the before and after amounts. If the step brought a positive amount to zero without setting a more specific prevention reason, `DamagePreventionReason.PipelineReducedToZero` is set automatically.

The package includes five built-in `DamageStep` implementations:

ApplyCriticalMultiplierStep

Critical hits deal amplified damage. This step multiplies the current damage amount by the critical hit multiplier when the incoming damage has been flagged as a critical strike. Its position in the pipeline is a design decision: placed before defensive steps, the multiplier scales raw (or modifier-adjusted) damage; placed after, it scales the already-mitigated result. Usually critical multipliers are applied early or as first step.

Scales the current damage by the critical hit multiplier when the damage is flagged as a critical hit. The step reads `DamageInfo.IsCritical` and `DamageInfo.CriticalMultiplier`. It is a no-op if `IsCritical` is `false`, or if the multiplier is ≤ 0 or exactly 1.0.

Both the critical flag and the multiplier are set in the `PreDamageContext` when building the damage request. A multiplier of 2.0 doubles the `Current` amount at the point where this step executes.

NOTE

The position of `ApplyCriticalMultiplierStep` in the pipeline determines what value the multiplier is applied to. Placing it before `ApplyDefenseStep` amplifies damage prior to defensive mitigation; placing it after scales post-defense damage. Design your strategy order accordingly.

ApplyDefenseStep

This step applies the target entity's defensive stat for this damage type — the primary stat-based mitigation layer in a typical RPG setup. For example, for a Physical damage type with Armor as its **Defensive Stat**, this step reads the target's Armor value, optionally reduces it by any armor penetration, then feeds the result into the **Damage Reduction Fn** to compute the mitigated damage. This is usually the step that eliminates most of the incoming damage for well-armored targets.

Applies defensive stat-based damage reduction as configured in the `DamageTypeSO`. The step reads the **Defensive Stat** and **Damage Reduction Fn**, and optionally the **Defense Penetration Stat** and **Defense Reduction Fn** (see [Defense Penetration](#)).

The step is a no-op when both **Defensive Stat** and **Damage Reduction Fn** are unset — which is the intended configuration for a damage type that should never be mitigated by defenses. If the configuration is inconsistent (one field set, the other null), a warning is logged and the step is skipped. The effective defensive value is computed after applying any penetration reduction, then fed into the **Damage Reduction Fn** to yield the final reduced amount. If the result is ≤ 0 , `DamagePreventionReason.DefenseAbsorbed` is set.

ApplyBarrierStep

Barriers act as a temporary shield layer in front of an entity's health pool: incoming damage hits the barrier first, and only what the barrier cannot absorb continues on to the next step. This step is the mechanism behind shielding abilities and any temporary protective layer you implement with the barrier system. Its position in the pipeline determines how much barrier is actually consumed: placing it after `ApplyDefenseStep` means the barrier only ever sees damage that already survived defensive mitigation, so the barrier is drained more slowly. Placing it before defensive steps causes the barrier to absorb the full, unreduced hit — making it a purer shield but one that depletes faster.

Consumes the target entity's barrier (temporary shield) to reduce incoming damage before it reaches health. If the `DamageTypeSO` has **Ignore Barrier** enabled, this step is skipped entirely. If the target has no `EntityHealth` component or its barrier is zero, the step is also a no-op.

The step subtracts from `DamageInfo.Amounts.Current` the portion absorbed by the barrier, and reduces the entity's barrier by the same amount. If the barrier fully absorbs the hit,

`DamagePreventionReason.BarrierAbsorbed` is set and the pipeline terminates. Damage that exceeds the barrier continues through subsequent steps as the updated `Current` value.

NOTE

`ApplyBarrierStep` does not reduce health directly. Any remaining damage after barrier absorption continues through the pipeline and is applied to health at the end of `TakeDamage`.

ApplyPercentageDmgModifiersStep

Percentage modifiers scale the incoming damage by a fraction. The three modifier layers (generic, source-specific, type-specific) each correspond to a stat whose value the target entity holds at runtime. A negative value reduces damage, a positive value amplifies it. When a single modifier layer reaches -100% or below on its own, the entity becomes immune to that entire category of damage for this hit.

Applies percentage-based damage modifiers from up to three layers, in order:

1. **Generic** — reads `AstraRpgHealthConfigSO.GenericPercentageDamageModificationStat` from the target's stats. Skipped if the `DamageTypeSO` has **Ignore Generic Percentage Modifiers** enabled.
2. **DamageSource-specific** — reads `DamageSourceSO.PercentageDamageModificationStat` from the target's stats.
3. **DamageType-specific** — reads `DamageTypeSO.PercentageDamageModificationStat` from the target's stats.

Each layer is evaluated individually for full immunity before contributions are summed. If the generic layer alone reaches $\leq -100\%$, `DamagePreventionReason.AllDamageImmune` is set and the step exits immediately. If the source-specific layer alone reaches $\leq -100\%$, `DamagePreventionReason.DamageSourceImmune` is set. If the type-specific layer alone reaches $\leq -100\%$, `DamagePreventionReason.DamageTypeImmune` is set. If no single layer triggers immunity, the three contributions are summed additively into a single net percentage applied to `Current` in one operation.

ApplyFlatDmgModifiersStep

Where percentage modifiers scale damage proportionally, flat modifiers add or subtract a fixed amount regardless of the hit's magnitude. A -15 flat absorb always removes 15 damage, whether the incoming hit was 20 or 2000. Flat modifiers are typically placed after percentage modifiers so they adjust the already-scaled value — but you are free to order them as your game design requires.

Applies flat damage modifiers from the same three layers as `ApplyPercentageDmgModifiersStep`:

1. **Generic** — reads `AstraRpgHealthConfigSO.GenericFlatDamageModificationStat`. Skipped if the `DamageTypeSO` has **Ignore Generic Flat Modifiers** enabled.
2. **DamageSource-specific** — reads `DamageSourceSO.FlatDamageModificationStat`.

3. **DamageType-specific** — reads `DamageTypeSO.FlatDamageModificationStat`.

All three contributions are summed additively. The net value is added to or subtracted from `DamageInfo.Amounts.Current`. The result is clamped to a minimum of 0: flat modifiers cannot bring damage below zero.

NOTE

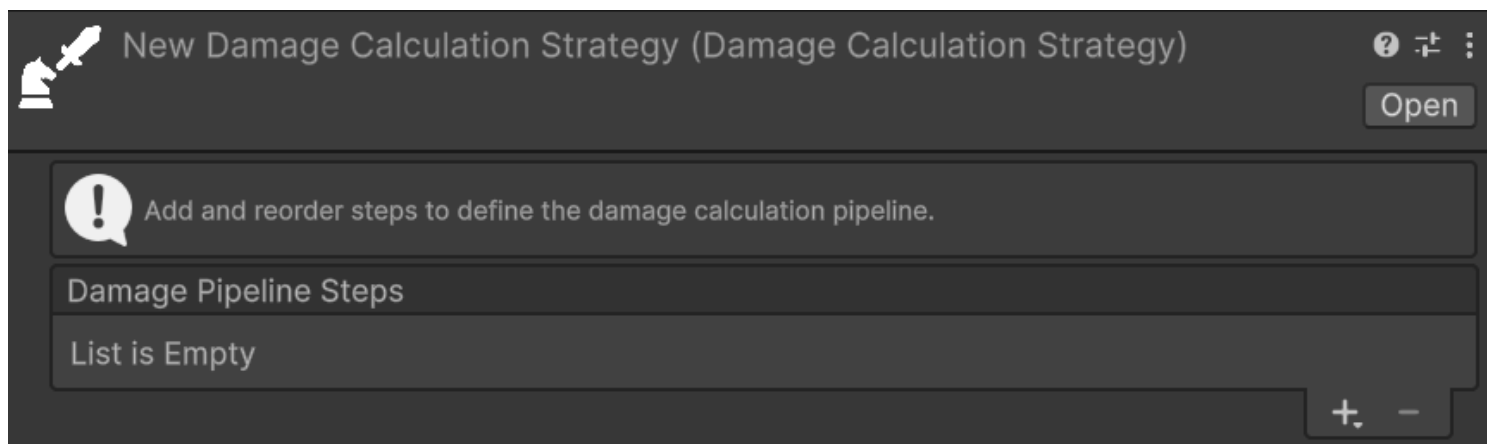
The conventional order is to run percentage modifiers before flat modifiers, so that flat additions or reductions are applied to the already-scaled value. This ordering is not enforced by the system; you can arrange steps freely.

Damage Calculation Strategy

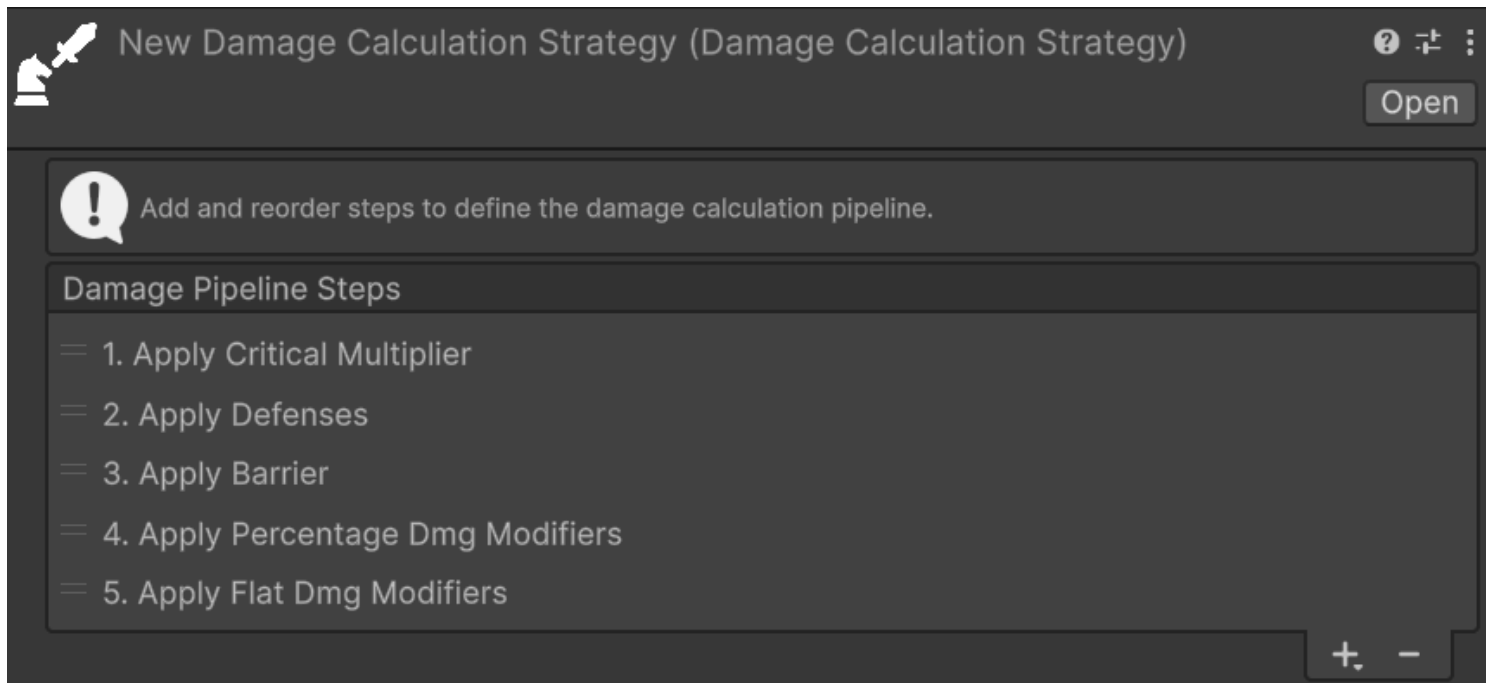
Relative path: `Damage Calculation Strategy`

Rather than embedding damage calculation logic in scripts, Astra RPG Health externalises the pipeline into a `DamageCalculationStrategy` asset — a `ScriptableObject` you control entirely from the Inspector. Each strategy holds an ordered list of steps; changing the list changes the pipeline. This separation lets you design, tune, and swap calculation behaviours without touching any code. A boss encounter can use a custom strategy with a specific subset of steps (e.g., one final extra step that limits the incoming damage to a maximum value of 10% of the Max HP); regular enemies fall back to the project-wide default; a runtime debuff can temporarily replace an entity's strategy to alter how it takes damage during a special encounter phase (e.g., all lightning damage is doubled).

In the Inspector, the strategy exposes a reorderable list labelled **Damage Pipeline Steps**. Steps can be added via a type-selection dropdown (the "Step" suffix is trimmed from display names for readability), removed, and reordered via drag. A newly created, empty strategy looks like this:



Once steps are added and arranged, the Inspector shows them as a numbered list reflecting the execution order:



Every `EntityHealth` component resolves the active strategy at runtime through a **three-tier priority**:

1. **Override Damage Calculation Strategy** — takes precedence over all other settings. Intended for temporary runtime effects (e.g., a debuff that temporarily changes how an entity takes damage) or for testing when a custom strategy is already assigned.
2. **Custom Damage Calculation Strategy** — an entity-specific strategy that overrides the global default but can itself be overridden at runtime by the tier above.
3. **Default Damage Calculation Strategy** — configured in `AstraRpgHealthConfigSO` and applied to every entity that defines neither a custom nor an override strategy.

⊗ **IMPORTANT**

If no strategy is resolved — neither custom nor override is set on the entity, and no default is configured in `AstraRpgHealthConfigSO` — `TakeDamage` logs an error and the pipeline does not run.

Healing

An entity can be healed in 4 different ways:

- Direct healing through the `Heal` method.
- Health regeneration. This can be passive, as defined in the [Health Regeneration](#) configuration, so automatically triggered by the framework every so often, or it can be manually activated via the `ManualHealthRegenerationTick` method. Also for the latter, you must configure the statistic to consider for the calculation of healing through the configuration.
- Through lifesteal, that is healing the entity based on the damage dealt. However, lifesteal functioning is detailed in [Lifesteal](#) as it deserves a dedicated section.
- Via resurrection with the two `Resurrect` methods. One to resurrect the entity with a percentage of HP, and one to resurrect it with a fixed amount of health points. Applicable only if the entity is dead. Also resurrection is detailed in a dedicated section, [Resurrection](#).

NOTE

Regeneration (both passive and manual), lifesteal, and resurrection all use, behind the scenes, the `Heal` method to effectively heal the entity.

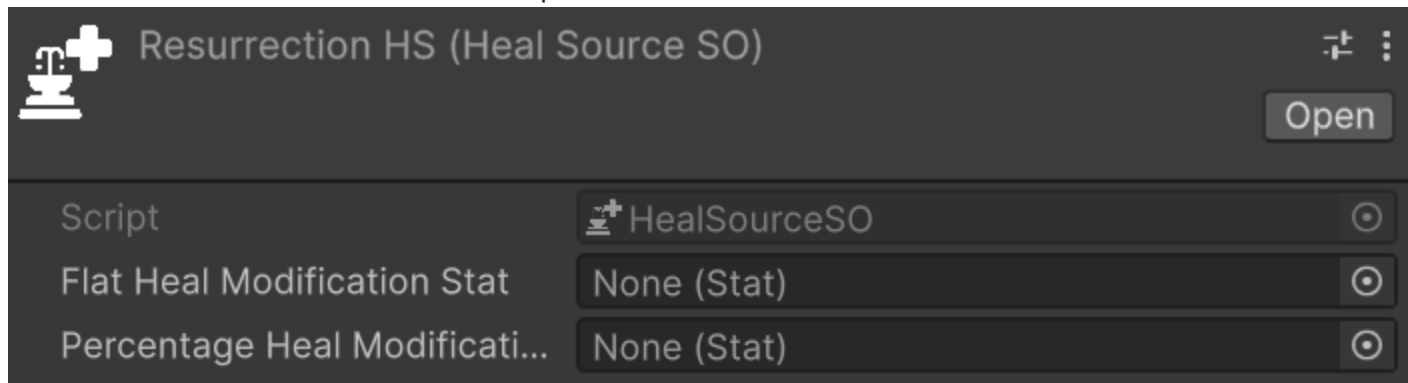
We will first introduce some concepts that are common to all the four healing methods, and then we will analyze direct healing and health regeneration. As aforesaid, lifesteal and resurrection are analyzed in their respective sections, [Lifesteal](#) and [Resurrection](#).

Heal Source

Relative path: `Heal Source`

A `HealSourceSO` represents the source of healing. Some examples of `HealSourceSO` could be: healing potions, skills, passive regeneration, lifesteal, system (e.g., the player is healed to max health during the tutorial by a support machine), and so on. The `HealSourceSO` is an important concept as it allows distinguishing between different sources of healing, and thus applying specific healing modifiers for each healing source, in addition to the generic healing modifiers that apply to all healing sources. Furthermore, `HealSources` are important because they are communicated in healing contexts, and therefore in the listeners of healing events, allowing the latter to react specifically based on the healing source that triggered the event. Thanks to this, you could, for example, implement a buff that increases the healing received from healing potions. Finally, `HealSources` pave the way for the implementation of a combat log that tracks not only the amount of healing received but also the healing source, allowing the player to have a deeper understanding of what healed them most effectively in a specific game situation.

An instance of `HealSourceSO` in the inspector looks like this:



You can notice that there are two properties to configure for each `HealSourceSO`:

- **Flat Heal Modification Stat:** the statistic to consider in an entity to apply flat, specific healing modifiers for this healing source. Positive values of this statistic increase the healing received from this source, while negative values decrease it. If the entity does not have this statistic, a value of 0 will be considered for this statistic, and therefore no specific flat healing modifier will be applied for this healing source.
- **Percentage Heal Modification Stat:** the statistic to consider in an entity to apply percentage, specific healing modifiers for this healing source. Positive values of this statistic increase the healing received from this source, while negative values decrease it. If the entity does not have this statistic, a value of 0 will be considered for this statistic, and therefore no specific percentage healing modifier will be applied for this healing source.

Heal Modifiers

Heal modifiers are statistics that influence the amount of healing received by an entity. They can influence it positively or negatively, depending on the value. There are two types of healing modifiers: generic heal modifiers, which apply to all healing sources, and source-specific heal modifiers, which apply only to a specific healing source. Furthermore, healing modifiers can be flat or percentage-based. Flat healing modifiers influence the amount of healing received by adding or subtracting a fixed amount of HP to the healing, while percentage healing modifiers influence the amount of healing received by multiplying the healing by a certain factor. It is important to note that flat healing modifiers are applied before percentage healing modifiers. Therefore, percentage modifications will also affect the impact of flat healing modifiers.

For percentage modifiers, a statistic with a value of 120 increases the healing by 120%, while a statistic with a value of -25 decreases the healing by 25%.

Generic healing modifiers are summed with the source-specific ones: flat modifiers are added together, as are the percentage ones.

Finally, if an entity does not possess the statistic associated with a healing modifier in its Stat Set, a value of 0 will be considered for that statistic, and therefore no healing modifier will be applied for that

statistic.

Clearly, to provide healing modifiers to entities, since they are statistics, you must proceed through the Stat Modifiers APIs, as described in the [Understanding Stat Modifier Types](#) section of the base framework documentation.

Example of Heal Modifiers

Suppose we want to calculate the healing received by an entity using a healing potion. Here are the example values:

Definition of the statistics used in the example:

- **Potion Heal Flat Modifier:** Source-specific statistic (the potion) that adds or subtracts a fixed value to the HP healed by the potion.
- **Potion Heal Percentage Modifier:** Source-specific statistic (the potion) that increases or decreases the potion's healing by a percentage.
- **Generic Flat Heal Modifier:** Generic statistic that applies to all healing sources, adding or subtracting a fixed value to the HP healed.
- **Generic Percentage Heal Modifier:** Generic statistic that applies to all healing sources, increasing or decreasing the healing by a percentage.

Phase	Value	Calculation	Result
Base healing	80 HP	-	80 HP
Flat modifiers	Potion Heal Flat Modifier: 50 Generic Flat Heal Modifier: -10	$50 - 10 = 40$ HP $80 + 40$	120 HP
Percentage modifiers	Potion Heal Percentage Modifier: 20% Generic Percentage Heal Modifier: 10%	$20\% + 10\% = 30\%$ 120×1.3	156 HP

Final healing received: 156 HP

This table clearly shows how flat and percentage modifiers are summed, and how the final result is reached.

Direct Healing

The `Heal` method is the most straightforward way to heal an entity. It takes a `PreHealContext` as input, which contains all relevant information about the healing you intend to apply and operates as follows:

1. It checks if the entity is dead. If the entity is dead, will raise an exception, as you cannot heal a dead entity. If you want to resurrect a dead entity, check the [Resurrection](#) section.

2. It checks if the healing to be applied is marked as a critical hit and, if so it applies the critical multiplier to the healing amount.
3. It retrieves the generic and `HealSourceSO`-specific **flat** heal modifiers for the entity.
4. It retrieves the generic and `HealSourceSO`-specific **percentage** heal modifiers for the entity.
5. It retrieves the heal flat modifiers specific to the heal source, and sums them with the generic flat heal modifiers.
6. It retrieves the heal percentage modifiers specific to the heal source, and sums them with the generic percentage heal modifiers.
7. It applies the flat modifiers to the base healing amount, and then applies the percentage modifiers to the result, to calculate the final healing amount.
8. It increases the entity's current HP by the final healing amount.
9. It raises the *Entity Healed Event* with a `HealResolutionContext` containing all relevant information about the healing that was just applied.
10. It returns the `HealResolutionContext` created in the previous step so that the caller can use it for further processing if needed.

The `Heal` method, differently from the `TakeDamage` method, cannot be configured to reorder the steps of the healing pipeline. This choice was made as healing is generally more straightforward than damage, and there are usually no mechanics that require reordering the steps of the healing pipeline. Therefore, I evaluated simplicity and ease of use more important than flexibility for this method.

Heal Usage Example

Suppose we want to heal an entity for 20% of its total max HP. The code could be the following:

```
// Assuming that:  
// - healSource is a HealSource representing the healing coming from a skill  
// - skillCaster is the EntityCore that casts the healing skill  
// - target is the EntityCore that we want to heal
```

```
if (target.TryGetComponent(out EntityHealth targetHealth)) {  
    // in case the target has a EntityHealth component...  
    long healAmount = target.GetMaxHpPortion(0.2d);
```

```
    PreHealContext preHealContext = PreHealContext.Builder  
        .WithAmount(healAmount)  
        .WithSource(healSource)  
        .WithHealer(skillCaster)  
        .WithTarget(target.EntityCore)  
        .Build();
```

```
    targetHealth.Heal(preHealContext);  
}
```

It is not a good practice to hardcode the healing amount directly in code (20% of max hp in the example). Hardcoded values make tuning and balancing difficult, prevent reuse across different skills or items, and force code changes and recompilation for values that designers or content creators should be able to tweak. Prefer one of these approaches instead:

- **Serialize the amount** as an inspector-editable field so designers can tweak values without touching code.
- **Use a `ScalingFormula`** (or equivalent) to compute the heal amount dynamically from the caster's or target's stats, allowing the effect to scale with progression and remain data-driven.

Example using a `ScalingFormula` to calculate the heal amount at runtime:

```
// Assuming that scalingFormula is a ScalingFormula that calculates the heal amount
// based on the skill caster's stats...
long healAmount = scalingFormula.CalculateValue(skillCaster);
```

```
PreHealContext preHealContext = PreHealContext.Builder
    .WithAmount(healAmount)
    .WithSource(healSource)
    .WithHealer(skillCaster)
    .WithTarget(target.EntityCore)
    .Build();
```

```
targetHealth.Heal(preHealContext);
```

Recommended pattern when healing programmatically:

1. Construct a `PreHealContext` with all relevant information using the fluent builder.
2. Call `Heal` passing the constructed context.

The `PreHealContext` fluent builder is helpful because it guides you through required inputs first — the IDE will typically present required builder steps one at a time. When the builder presents multiple fields together, those fields are optional and can be omitted.

Health Regeneration

Health regeneration is a mechanism that allows an entity to passively regenerate its health over time. In games generally, this mechanic is implemented through discrete regeneration ticks. Ticks, depending on the game, can be marked by the passage of time, or by certain events, like the end of a turn in a turn-based game.

This package supports both passive and manual regeneration.

Passive Health Regeneration

Passive regeneration, to work, needs to be configured through the [Health Regeneration](#) section of the package configuration. The following parameters should be configured there:

- **Health Regeneration Source:** the `HealSourceSO` to associate with regeneration (passive and manual). The package will use this `HealSourceSO` to create the healing contexts to pass to the `Heal` method when it's time to regenerate health.
- **Passive Health Regeneration Stat (HP/10s):** the amount of HP the entity regenerates every 10 seconds. The package will take care of scaling this amount based on the time that passes between consecutive ticks, to ensure smooth and consistent regeneration.
- **Passive Health Regeneration Interval:** the time interval, in seconds, between one regeneration tick and the next.

Additionally, it is important to remember to enable passive regeneration on the entity's `EntityHealth`.

To give an example, if we want an entity to regenerate 5 HP every 0.5 seconds, we must configure the "Passive Health Regeneration Interval" parameter to 0.5, and the entity must have the statistic associated with "Passive Health Regeneration Stat (HP/10s)" set to a value of 100 (5 HP every 0.5 seconds correspond to 10 HP per second, which correspond to **100 HP every 10 seconds**).

Performance Considerations

Before making any considerations about performance, I want to remind you that premature optimization is the root of all evil. Before taking actions to optimize performance, a real performance problem should be identified through profiling. That said, since by default passive regeneration raises Entity Healed Events (global and extra), it's important to consider the performance impact if you have a large number of entities regenerating health simultaneously with a very short regeneration interval. If there were also several listeners subscribed to the global Entity Healed Event, you would get several function calls at each regeneration tick X each entity. If this becomes a performance issue for your game, you have several levers available to optimize performance:

- **Increase the regeneration interval:** by increasing the regeneration interval, the number of regeneration ticks per time unit is reduced, and therefore the number of Entity Healed Events raised.
- **Disable the raising of Entity Healed Events for passive regeneration:** if you don't need to react to passive regeneration through the listeners of Entity Healed Events, you can disable the raising of these events for passive regeneration through the configuration. This way, even if you have a large number of entities regenerating health simultaneously with a very short regeneration interval, you won't have a significant performance impact due to raised events and associated function calls.
- **Use Extra Entity Healed Events:** if you only need to react to the passive regeneration of one or some specific entities, you can configure these entities to raise an Extra Entity Healed Event specifically for passive regeneration, and subscribe your listeners to this event instead of the global Entity Healed Event. This way, the only function calls you'll have at each tick will be those of the listeners subscribed to the specific Extra Entity Healed Event of the entities configured to raise it, thus reducing the performance impact.

I would like to reiterate that, unless you have identified a real performance problem through profiling, it is not necessary to take any of these actions to optimize performance. You probably won't have any performance issues even with a thousand entities and dozens of listeners subscribed to the global Entity Healed Event.

Manual Health Regeneration

Manual regeneration is instead triggered manually through the [ManualHealthRegenerationTick\(\)](#) API method.

This method, when called, triggers a regeneration tick for the entity. Note that the statistic considered for the health calculation does NOT coincide with the one configured for passive regeneration, but is a separate statistic, also set via the package configuration, specifically: [Manual Health Regeneration Stat](#). The value of this statistic determines the amount of HP regenerated at each manual regeneration tick. Obviously, healing modifiers, both generic and those specific to the heal source configured for regeneration, will also affect manual regeneration, just like passive regeneration.

Passive vs Manual Health Regeneration: Which One Should I Use?

If you have doubts about which type of regeneration to use, here are some guidelines that might help you choose:

- **Real Time:** if your game is in real-time, passive regeneration is generally more suitable, as it integrates better with continuous gameplay flow.
- **Turn-Based:** if your game is turn-based, manual regeneration is generally more suitable, as you can trigger it at the end of each turn, ensuring more precise control over the game flow.
- **Regeneration in the base:** if your game involves a regeneration mechanic that only happens when the character is in a specific location (e.g., a base), you can use passive regeneration, but activate it only when the character is in that location, and deactivate it when they leave. If, on the other hand, you want it to be active even when the character is outside the base, but still want it to be stronger when in the base, you can assign a specific Heal Modifier for the regeneration's Heal Source when the character is in the base, to increase the amount of HP regenerated when the character is at the base. Upon leaving the base, you can remove this Heal Modifier to reduce the amount of HP regenerated.
- **Regeneration out of combat:** if your game features regeneration that only happens when the character is out of combat, you can use passive regeneration, turning it on when out of combat, and turning it off when entering combat. Or, if you want it to be active during combat as well, but stronger out of combat, you can (as with base regeneration) add a specific Heal Modifier for the regeneration's Heal Source when the character is out of combat.
- **Regeneration as a buff:** if the entity receives a buff providing HP regeneration according to a specific timing, you can use manual regeneration to trigger regeneration ticks perfectly aligned with

the timing intended by the buff.

Clearly, these are just guidelines, not strict rules. For every problem, there are always several possible solutions, and choosing the best one always depends on the specific context of your game and the mechanics you want to implement. So, if you feel a solution different from the suggested ones fits your game better, feel free to use it.

Lifesteal

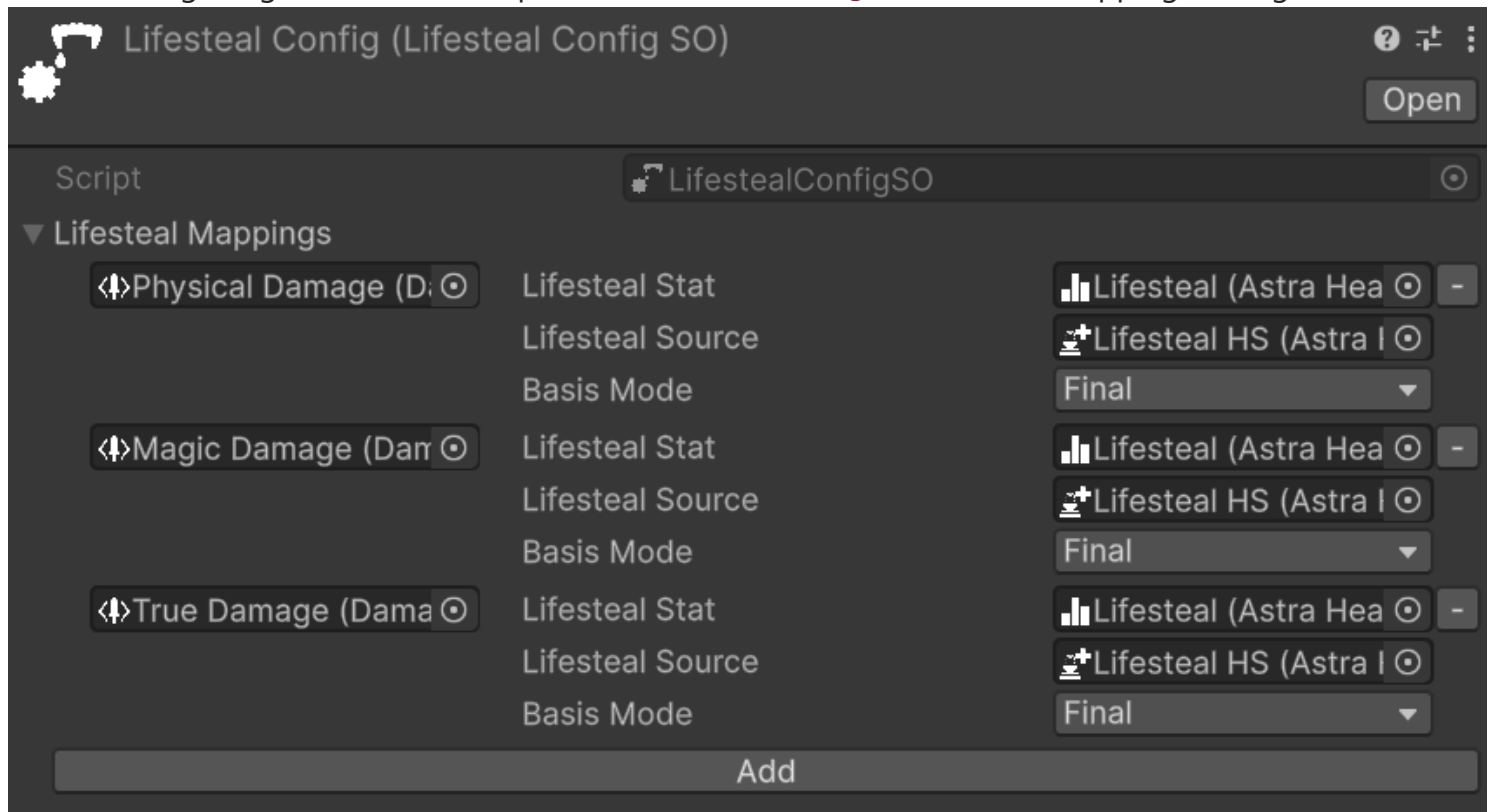
Lifesteal is a mechanic that lets an entity recover health proportional to the damage it deals. When an entity successfully damages a target, a percentage of the damage — determined by a configurable stat — is healed back to the attacker. The lifesteal percentage, the heal source used, and which point in the damage pipeline is sampled as the heal basis are all configurable per damage type, giving designers fine-grained control without any additional code.

Lifesteal Config

Relative path: **Astra RPG Health/Lifesteal Config**

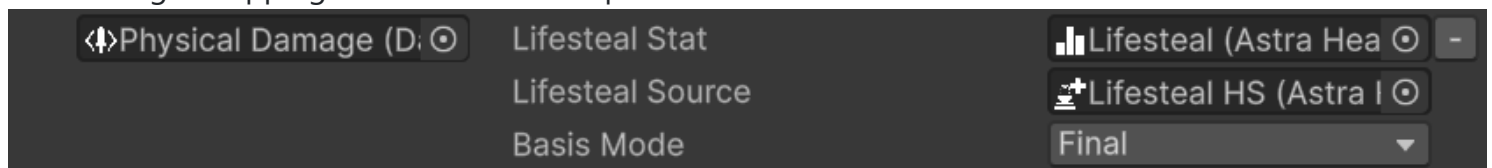
The **LifestealConfigSO** is the root asset for lifesteal configuration. It holds a dictionary that maps each **DamageTypeSO** to a **LifestealStatConfig**, defining how lifesteal behaves for that specific damage type. Damage types not present in the dictionary do not trigger lifesteal.

The following image shows an example of a **LifestealConfigSO** with three mappings configured:



Configuring Lifesteal Mappings

Each entry in the dictionary pairs a **DamageTypeSO** with a **LifestealStatConfig**. The following image shows what a single mapping looks like in the inspector:



There are three fields to configure for each mapping:

Lifesteal Stat

The stat that drives the lifesteal percentage. At runtime, the attacker reads this stat value from its own `StatSet` and computes the heal as:

$$\text{heal amount} = \text{basis damage} \times (\text{lifesteal stat value} / 100)$$

The stat value is read as a `Percentage` type, meaning the raw `long` value stored in the stat is divided by 100 internally. A stat value of `15` therefore corresponds to 15% lifesteal. For lifesteal to activate, this stat must be present in the dealing entity's `StatSet`.

Lifesteal Source

The `HealSourceSO` used when applying the lifesteal heal. As with any heal in the framework, the heal source determines which heal modifiers — flat and percentage — are applied to the resulting heal, so lifesteal heals can be boosted or reduced by modifiers associated with the configured `HealSourceSO`. See [Heal Source](#) for details on `HealSourceSO` configuration.

Amount Selector

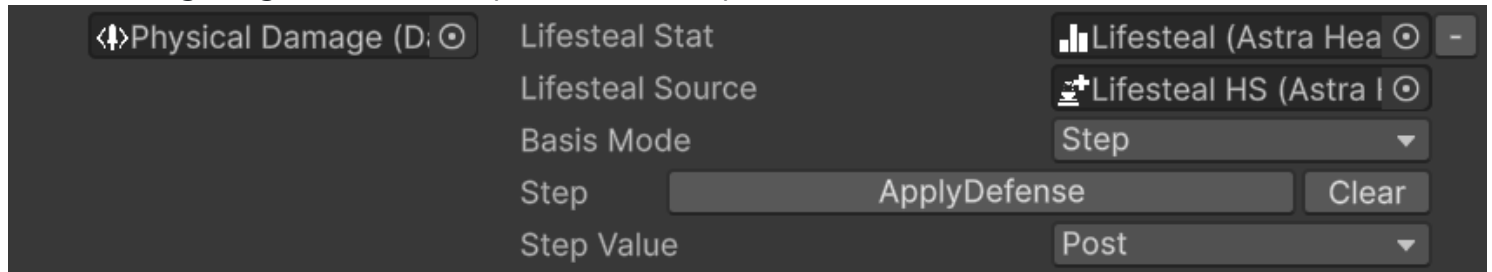
Determines which damage value is used as the basis for the lifesteal computation. See [Amount Selector](#) below.

Amount Selector

The Amount Selector controls which damage value is sampled as the lifesteal basis. Three modes are available:

- **Final** (*default*): uses the damage value after all pipeline steps — the damage actually applied to the target. This is the most intuitive mode for a straightforward "heal for a percentage of damage dealt" mechanic.
- **Initial**: uses the raw damage value before any pipeline modifications. Useful when you want lifesteal to reflect the attacker's power output regardless of the target's defenses — for example, a vampire's lifesteal that scales with their attack power rather than with how much the target's armor reduced.
- **Step**: samples the damage value at a specific pipeline step. You select a step and whether to use the value recorded before (**Pre**) or after (**Post**) that step executes, giving designers precise control over which phase of the calculation drives the heal. Refer to [Damage Calculation Pipeline](#) for an overview of the available pipeline steps.

The following image shows the inspector when Step mode is selected:



NOTE

If **Step** mode is selected but no step is configured, the system falls back to **Final** damage. The inspector displays a warning to indicate this condition.

Adding Lifesteal to the Package Configuration

Once the `LifestealConfigSO` is set up, assign it to the **Lifesteal Config** field in the `AstraRpgHealthConfigSO`. All entities that share this configuration asset will have lifesteal enabled according to the mappings defined in the assigned `LifestealConfigSO`. See [Lifesteal](#) in the Package Configuration reference for field details.

Performance Considerations

NOTE

Before making any considerations about performance, keep in mind that premature optimization is the root of all evil. A real performance problem should be identified through profiling before taking any action.

By default, lifesteal heals raise the standard healing events (`Pre Heal` and `Entity Healed`). In games where many entities deal lifesteal damage frequently — for example, a horde of enemies all with a lifesteal stat, each landing hits several times per second — the volume of heal events generated can become significant, especially when the global `Entity Healed Event` has multiple listeners.

If profiling reveals this to be a bottleneck, the **Suppress Lifesteal Events** flag in `AstraRpgHealthConfigSO` disables healing event emission for lifesteal heals entirely. The heal is still applied; only the events are suppressed. This option is appropriate when you do not need to react to lifesteal heals through event listeners — for example, when your UI or combat log does not track lifesteal healing specifically.

A similar consideration applies to passive health regeneration, which can also generate a high volume of heal events under certain conditions. See [Performance Considerations](#) in the Healing documentation for

a broader discussion.

Conditions

NOTE

Lifesteal is evaluated after every damage application. The following conditions must all be met for it to trigger:

- The dealing entity is the source of the damage. Lifesteal does not trigger on damage dealt by others.
- The dealing entity is alive at the moment of damage resolution.
- `AstraRpgHealthConfigSO` has a `LifestealConfigSO` assigned.
- The `DamageTypeSO` of the hit has a mapping in the `LifestealConfigSO`.
- The configured **Lifesteal Stat** is present in the dealing entity's `StatSet`.
- The computed heal amount is greater than zero. A lifesteal stat of 0% produces no heal.

IMPORTANT

Lifesteal relies on the **Global Damage Resolution Event** being correctly configured on each entity's `EntityHealth`. This event is used by the framework to propagate damage outcomes across all listening entities — including the attacker, which uses it to detect its own successful hits and trigger lifesteal. Assigning a non-global event to this slot will break lifesteal and other framework features that depend on centralized damage communication. See [Global Events](#) for setup details.

Health Scaling Component

Relative path: `Scaling -> Health Scaling Component`

Base framework convenience relative path: `Astra RPG Framework -> Scaling -> Health Scaling Component`

`HealthScalingComponentSO` is a `ScalingComponent` provided by Astra RPG Health that calculates a scaled value from an entity's health metrics — current HP, maximum HP, or missing HP. It plugs into `ScalingFormula` assets as any other scaling component, enabling mechanics such as HP-based damage abilities and shields or heals that grow proportionally to an entity's health pool.

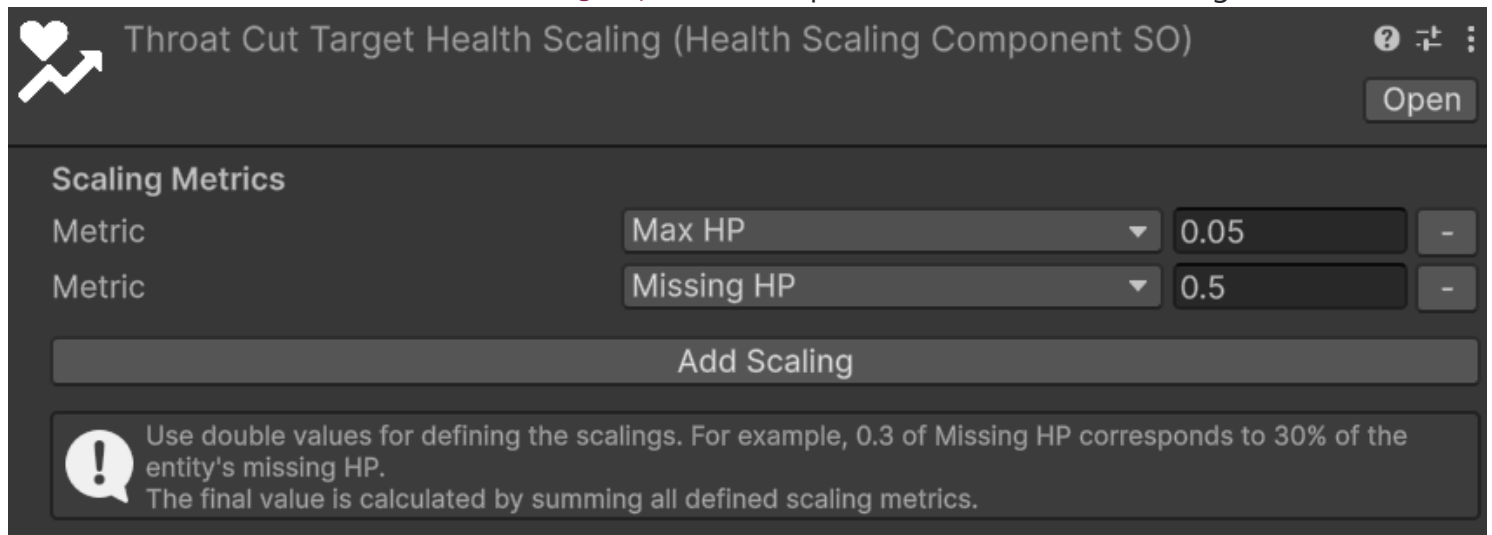
NOTE

`HealthScalingComponentSO` requires the target entity to have an `EntityHealth` component. If the component is absent, `CalculateValue` returns 0 and logs a warning. Refer to [Entity Health Component](#) for setup details.

Inspector

The inspector exposes a **Scaling Metrics** list, where each entry pairs one health metric with a `double` multiplier.

Here is a screenshot of the `HealthScalingComponentSO` inspector with two entries configured:



Each entry consists of:

- **Metric**: the health value to scale on. See [Health Metrics](#) for the available options.
- **Value** (*double*): the multiplier applied to the chosen metric. `0.3` means 30%.

Use the **Add Scaling** button to append a new entry and the – button on the right of each row to remove it.

Health Metrics

Metric	Description
HP	The entity's current HP at the moment of calculation.
Max HP	The entity's current maximum HP.
Missing HP	The difference between maximum and current HP (<code>max - current</code>).

Combining Multiple Metrics

A single `HealthScalingComponentSO` can contain any number of entries. The final value is the integer sum of each individual contribution:

```
total = round(metric1_value × multiplier1) + round(metric2_value × multiplier2) + ...
```

For example, a component configured with `Missing HP: 0.3` and `Max HP: 0.1` on an entity with 200 Max HP and 80 current HP (120 missing) yields:

```
round(120 × 0.3) + round(200 × 0.1) = 36 + 20 = 56
```

Usage in a Scaling Formula

`HealthScalingComponentSO` is a `ScalingComponent`, so it plugs directly into the **Self Scaling Components** or **Target Scaling Components** lists of a `ScalingFormula`. Refer to the [Scaling Formulas](#) section of the base framework documentation for the full workflow.

TIP

Assign the component to **Target Scaling Components** when the value should reflect the *target's* health state — for example, execute damage that scales with the target's missing HP. Use **Self Scaling Components** when the value reflects the *caster's* state — for example, a shield whose strength scales with the caster's max HP.

Experience Collection

Experience collection is the mechanism through which entities earn experience points when they contribute to enemy kills (or entities deaths in general in some cases). In Astra RPG Health, the system revolves around two actors: the *collector*, the entity that receives experience, and the *victim*, the entity that dies and provides it.

When a victim dies, its `EntityHealth` raises the [Global Entity Died](#) event. Any `ExpCollector` component listening to that event evaluates its configured strategy to determine whether XP should be awarded — and if so, extracts the XP amount from the victim and forwards it to the collector's `EntityLevel`. The entire validation and collection logic lives inside a dedicated *Experience Collection Strategy* asset, making it straightforward to swap, extend, or compose strategies without touching any component code.

⊗ IMPORTANT

Experience collection relies on the **Global Entity Died** event configured in your `AstraRpgHealthConfigSO`. Ensure this event is assigned before entering Play Mode — without it, `ExpCollector` will never be notified of kills. See [Global Entity Died Event](#) for details.

Prerequisites

Before setting up experience collection, ensure the following are in place.

On each **victim** entity:

- **The victim must expose experience through an `ExpSource` component.** The Astra RPG Framework provides a concrete `ExpSource : MonoBehaviour` that stores the XP amount and a harvested flag. Attach this component to any entity that should provide XP when it dies, and configure the amount accordingly. Refer to the Framework documentation for details:

<https://electricdrill.github.io/AstraRpgFrameworkDocs/MD/workflows.html#exp-source> [↗](#)

On the **collector** entity:

- **The collector must be an entity.** Therefore, it must have the `EntityCore` component.

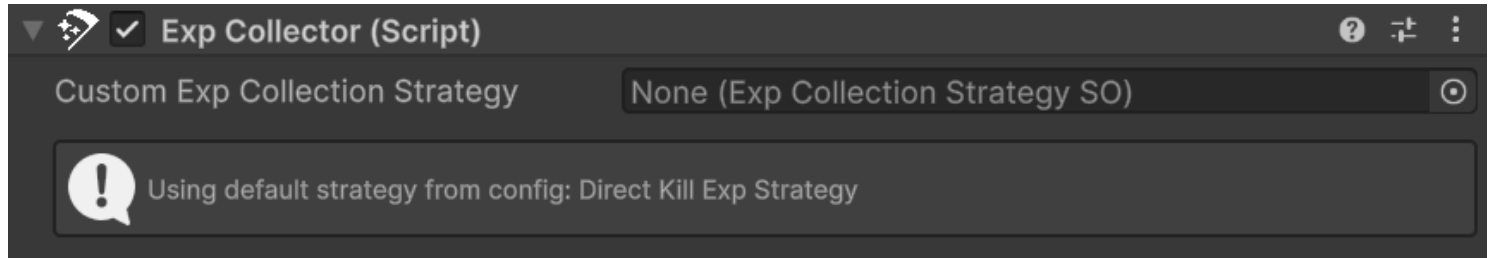
Setting Up the Exp Collector

`ExpCollector` is the component that bridges the global death event and the configured strategy. Add it to any entity that should receive XP from kills.

`ExpCollector` automatically subscribes to the [Global Entity Died Event](#) configured in `AstraRpgHealthConfigSO` — no manual event wiring is required. Whenever any entity dies, the

component evaluates its configured strategy to determine whether XP should be awarded.

The following image shows the `ExpCollector` inspector:



Custom Exp Collection Strategy is the optional per-entity override. Assign a strategy here to give this specific collector its own behavior — for example, a boss that earns XP only from specific kill types, or a trap system with its own award logic. When left empty, the collector falls back to the project-wide default configured in `AstraRpgHealthConfigSO` (see [Default Strategy in Package Configuration](#)).

Inspector Status Box

The custom `ExpCollector` inspector includes a status indicator that shows at a glance whether an active strategy is in place:

- **Info** — "Using custom strategy: [name]": a **Custom Exp Collection Strategy** is assigned on this component.
- **Info** — "Using default strategy from config: [name]": no custom strategy is assigned, but a default is found in the `AstraRpgHealthConfigSO`.
- **Warning** — "No AstraRpgHealthConfig found...": no package configuration asset is in scope. See [Package Configuration](#) for setup details.
- **Warning** — "No default experience collection strategy configured in AstraRpgHealthConfig...": a configuration asset exists but no **Default Exp Collection Strategy** is set on it.

Default Strategy in Package Configuration

Collectors without a **Custom Exp Collection Strategy** fall back to the **Default Exp Collection Strategy** set in the `AstraRpgHealthConfigSO` (see [Package Configuration - Experience](#)). Configuring this field once means every bare `ExpCollector` in the project inherits the same behavior without needing an explicit assignment on each one.

Strategy resolution at runtime follows this order:

1. **Custom strategy** assigned on the `ExpCollector` → used directly.
2. **No custom strategy** → default strategy from `AstraRpgHealthConfigSO` → used.
3. **No custom strategy and no config or default** → a warning is logged and no XP is collected.

Built-in Strategies

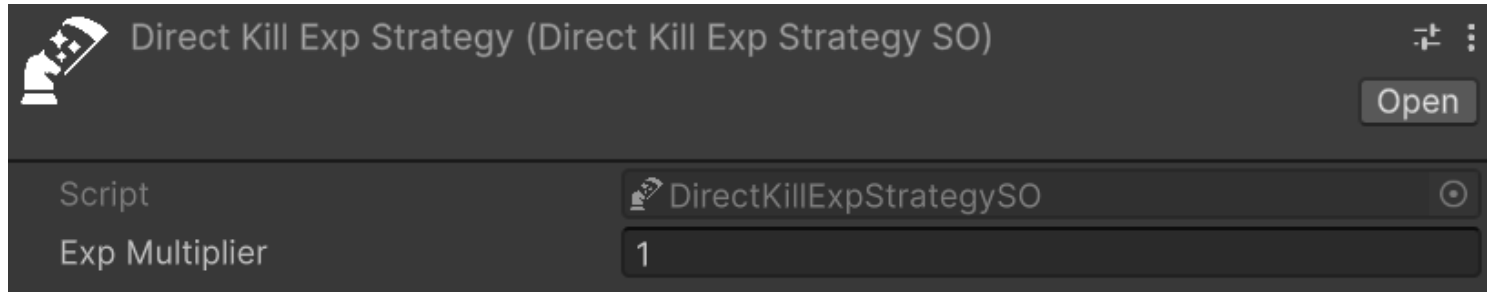
Three strategies are provided out of the box. Each is a `ScriptableObject` asset you create, configure, and assign to an `ExpCollector` or set as the project default in `AstraRpgHealthConfigSO`.

Direct Kill

Relative path: Astra RPG Health -> Exp Collection Strategies -> Direct Kill

`DirectKillExpStrategySO` is the most straightforward strategy: the collector receives XP only if it personally dealt the finishing blow. This is the natural choice for games where XP belongs exclusively to the entity that secured the kill.

The following image shows a `DirectKillExpStrategySO` in the inspector:



- **Exp Multiplier:** a float multiplier applied to the victim's base XP amount (default: `1.0`). Values greater than `1` grant a bonus — for example, `2.0` awards double XP for personal kills.

For XP to be awarded, all of the following must hold:

- The entity that dealt the final blow is the collector itself.
- The victim has an `ExpSource` component.
- The victim's `ExpSource` has not already been harvested by another collector.

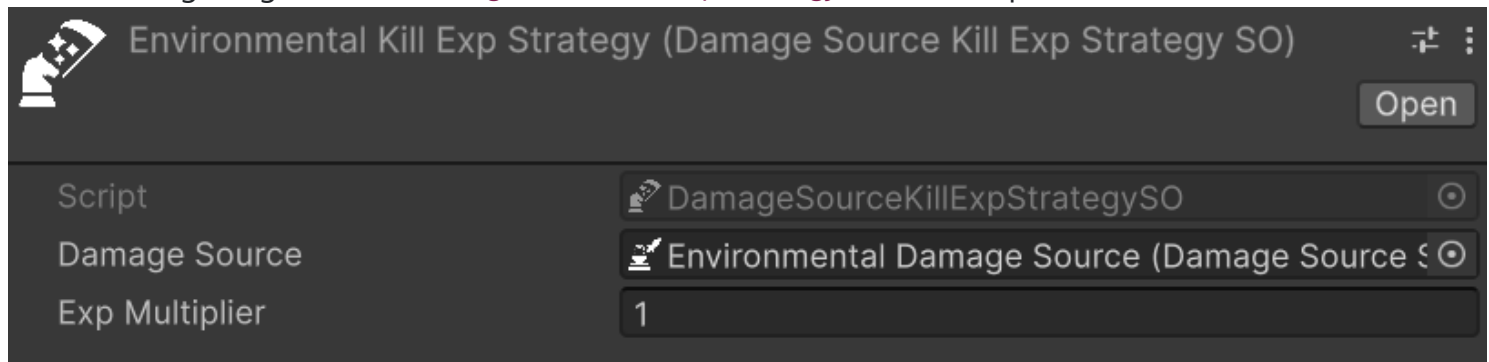
The harvested check ensures that a single kill cannot be claimed twice when multiple `ExpCollector` entities use non-harvested-dependent strategies.

Damage Source Kill

Relative path: Astra RPG Health -> Exp Collection Strategies -> Damage Source Kill

`DamageSourceKillExpStrategySO` grants XP to the collector whenever a kill is delivered by a specific `DamageSourceSO`, regardless of which entity caused it. This is the right strategy for scenarios where the *cause* of death — rather than the attacker — determines who is rewarded.

The following image shows a `DamageSourceKillExpStrategySO` in the inspector:



- **Damage Source** (*required*): the `DamageSourceSO` whose lethal hits trigger XP collection on this collector.
- **Exp Multiplier**: float multiplier on the victim's base XP (default: `1.0`).

Let's look at a classic example from the Souls-like genre. In those games, the player still earns souls when an enemy falls off a ledge and dies, even though the player did not deliver the final blow. To replicate this:

- Create a `DamageSourceSO` named "Environmental" and use it for all fall damage applied to enemies.
- Create a `DamageSourceKillExpStrategySO`, set **Damage Source** to "Environmental", and assign it to the player's `ExpCollector`.

Now whenever an enemy is killed by environmental damage, the player's collector fires and the player earns the XP.

NOTE

`DamageSourceKillExpStrategySO` does not check whether the victim's `ExpSource` has been harvested. If multiple `ExpCollector` entities use a strategy configured with the same `DamageSourceSO`, each of them receives XP independently when that source kills a victim. This is intentional for scenarios like the one above, but factor it into your design when multiple collectors may be active simultaneously.

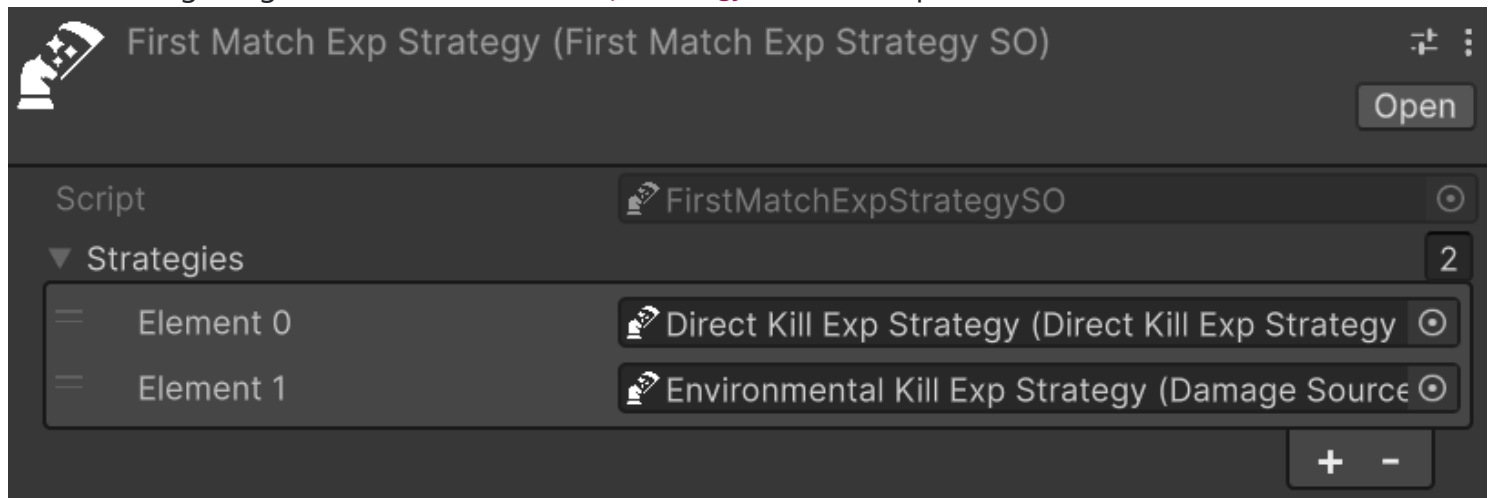
First Match Composite

Relative path: `Astra RPG Health -> Exp Collection Strategies -> First Match`

With the previous strategies, you are bound to a single condition for XP collection: either the collector must deal the killing blow, or the kill must come from a specific damage source. What if you want a more complex setup where multiple conditions can grant XP, each with its own priority?

`FirstMatchExpStrategySO` is a composite strategy that holds an ordered list of sub-strategies. It tries each in sequence and awards XP through the first one that succeeds, leaving the rest unevaluated.

The following image shows a `FirstMatchExpStrategySO` in the inspector:



- **Strategies:** the ordered list of `ExpCollectionStrategySO` assets to evaluate.

This is useful when an entity should earn XP under different conditions, each with a clear priority. For example:

- Try `DirectKillExpStrategySO` first — reward if the entity delivered the finishing blow.
- Fall back to `DamageSourceKillExpStrategySO` — reward if the kill came from a specific environmental source.

NOTE

Only the first matching sub-strategy collects XP. Once a strategy succeeds, the remaining entries in the list are skipped.

Custom Strategies

To implement experience collection logic beyond what the built-in strategies offer, extend `ExpCollectionStrategySO`. The class uses the Template Method pattern: `TryCollectExp` orchestrates the entire flow, and the virtual and abstract methods are the designated extension points.

The table below lists the available override points:

Method	Type	Purpose
<code>Validate</code>	abstract	Determine whether XP should be collected for this event. Returns a <code>multiplier</code> out parameter alongside the bool result.
<code>CollectExp</code>	virtual	Perform the actual collection: calculate the amount, call <code>AddExp</code> , and mark the source as harvested. Override for behaviors like distributing

Method	Type	Purpose
		XP across a party.
<code>CalculateExpAmount</code>	virtual	Compute the final XP amount from the source and the multiplier. Override to change the formula.
<code>TryGetExpSource</code>	virtual	Extract an <code>IExpSource</code> from the victim. Override if experience sources are not standard <code>MonoBehaviour</code> components on the victim's <code>GameObject</code> .
<code>TryCollectExp</code>	virtual	The full template method. Override for complete control over the collection flow, as <code>FirstMatchExpStrategySO</code> does.

For most custom strategies, overriding `Validate` is sufficient: implement your conditions, set the multiplier, and return `true` or `false`. Override `CollectExp` only when the collection action itself must differ — for example, to split the XP reward across multiple entities simultaneously.

For more information about the API and extension points, refer to the [ExpCollectionStrategySO](#) API documentation.

Resurrection

Resurrection brings a dead entity back to life, restoring it to a playable state with a configurable amount of HP. Unlike healing — which throws an exception when applied to a dead entity — resurrection is exclusively for entities that have already crossed their [death threshold](#).

The requested HP amount is a **base** value before modifiers. Internally, resurrection routes through the same `Heal` pipeline used for [direct healing](#), so all generic and `HealSourceSO`-specific heal modifiers apply. This means resurrection integrates naturally with any modifier system already in place: a buff that amplifies healing received will equally benefit resurrections, without any additional configuration.

Package Configuration

Before resurrection can be triggered, two fields in the `AstraRpgHealthConfigSO` need attention:

Default Resurrection Source

Type: `HealSourceSO` — **Required:** Yes

The `HealSourceSO` used automatically by the convenience `Resurrect` overloads and by the built-in [Resurrect Game Action](#) when no explicit source is provided. It categorises resurrection healing as a distinct type, enabling `HealSourceSO`-specific modifiers to apply. Without this field assigned, the convenience overloads fail at runtime with an assertion error.

See [Heal Source](#) for guidance on configuring `HealSourceSO` assets and [Default Resurrection Source](#) in the Package Configuration reference.

Default On Resurrection Game Action

Type: `GameAction<Component>` — **Required:** No

The [Game Action](#) executed automatically whenever any entity is resurrected, unless the entity defines its own [Override On Resurrection Game Action](#). Use this to implement a common post-resurrection behavior shared across all entities — playing a revival visual effect, re-enabling a disabled `GameObject`, or resetting specific state. Leave it empty if no shared behavior is needed.

See [Default On Resurrection Game Action](#) in the Package Configuration reference.

Resurrecting an Entity

`EntityHealth` implements `IResurrectable`, which exposes three overloads.

`Resurrect(PreHealContext)` gives full control over the resurrection: amount, source, and healer entity. It expects a `PreHealContext` built with the same fluent builder used in direct healing:

```

// Assuming:
// - resurrectionSource is the HealSourceSO categorising this resurrection
// - healer is the EntityCore of the character performing the resurrection (null for a
// system action)
// - target is the EntityCore of the dead entity to resurrect

if (target.TryGetComponent(out EntityHealth targetHealth)) {
    PreHealContext context = PreHealContext.Builder
        .WithAmount(targetHealth.MaxHp / 2)    // 50% of max HP as base amount,
before modifiers
        .WithSource(resurrectionSource)
        .WithHealer(healer)
        .WithTarget(target)
        .Build();

    targetHealth.Resurrect(context);
}

```

Refer to [Direct Healing](#) for a detailed explanation of `PreHealContext` and its fluent builder.

Resurrect(long) and **Resurrect(Percentage)** are convenience overloads that build the context automatically using the **Default Resurrection Source** from the configuration and a `null` healer (system-initiated resurrection). Use them when no specific caster is involved and the global default source is appropriate:

```

// Resurrect with a flat HP amount
targetHealth.Resurrect(100L);

// Resurrect with 30% of max HP
Percentage thirtyPercent = 30L;
targetHealth.Resurrect(thirtyPercent);

```

In all cases, the HP amount is the **base** value before heal modifiers are applied. Two constraints are always checked: the base amount must be strictly greater than the entity's death threshold, and must not exceed its maximum HP.

Heal Modifiers at Resurrection

Because resurrection calls `Heal` internally, the full modifier stack applies:

- **Generic flat heal modifiers** — add or subtract a flat HP amount from the base
- **Generic percentage heal modifiers** — scale the base HP up or down by a percentage

- **HealSourceSO-specific modifiers** — flat and percentage modifiers associated with the configured resurrection source

The order of application is the same as described in the [Direct Healing](#) section: first the flat modifiers (generic and **HealSourceSO**-specific) are summed and applied to the base amount, then all percentage modifiers (generic and **HealSourceSO**-specific) are summed and applied to the result.

The entity's HP after resurrection may therefore differ from the base amount requested.

`ResurrectionContext.NewValue` always reflects the **actual HP after all modifiers** have been applied.

NOTE

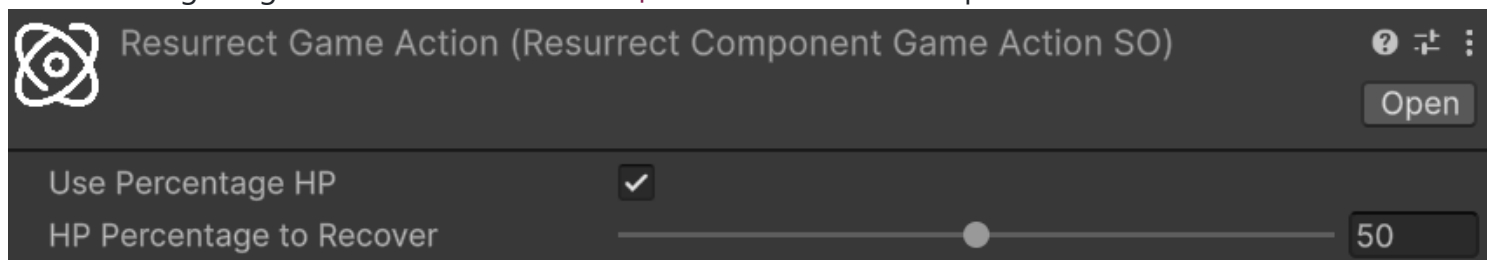
If active negative heal modifiers are strong enough to reduce the effective HP to or below the entity's death threshold, the system applies a fallback and forces HP to `death threshold + 1`, guaranteeing the entity is always resurrected alive. `ResurrectionContext.NewValue` reflects this clamped value.

The Resurrect Game Action

Relative path: `Astra RPG Framework/Game Actions/Context: Component/Resurrect`

`ResurrectComponentGameActionSO` is a ready-made [Game Action](#) that can be dropped into any Game Action slot in the inspector to resurrect an entity. It always uses the **Default Resurrection Source** from the configuration; no source field is exposed in the inspector.

The following image shows the `ResurrectComponentGameActionSO` inspector:



The configurable fields are:

- **Use Percentage HP**: if enabled, the entity is resurrected with a percentage of its maximum HP; if disabled, a flat HP amount is used instead
- **Percentage HP to Recover**: the percentage of max HP to restore, in the range 0–100. Active only when **Use Percentage HP** is enabled
- **Flat HP to Recover**: the flat HP amount to restore. Active only when **Use Percentage HP** is disabled

NOTE

If **Percentage HP to Recover** or **Flat HP to Recover** is set to 0, the game action falls back to resurrecting the entity with exactly 1 HP, guaranteeing the resurrection is always valid.

Common use case — automatic resurrection after a delay: combine `ResurrectComponentGameActionSO` with a [Delayed Game Action](#) inside the entity's **Override On Death Game Action** (or the global **Default On Death Game Action**). When the entity dies, the death game action fires a delayed sequence that triggers the `ResurrectComponentGameActionSO` after a configured number of seconds — implementing an automatic respawn without a single line of code. See the [Game Actions](#) documentation for details on composing and chaining game actions.

Per-Entity Customization

Each `EntityHealth` component exposes an **Override On Resurrection Game Action** field in its inspector that takes precedence over the **Default On Resurrection Game Action** defined in the configuration. Use it to give specific entities a unique post-resurrection behavior while keeping a sensible global default for everything else.

See [Death](#) in the Entity Health reference for the full list of per-entity death and resurrection inspector fields.

Resurrection Events

Resurrecting an entity raises multiple events. Because resurrection routes through `Heal`, the healing events fire first:

- **Pre Heal Event** and **Entity Healed Event** — raised by the internal `Heal` call, carrying the standard healing context before and after modifiers. Subscribe to these if your logic needs to react to the healing component of a resurrection; for example, to trigger a heal-triggered ability even during resurrection
- **Entity Gained Health Event** — raised when the entity's HP increases during the `Heal` call. Useful for general HP-change listeners that do not need the full context of the healing, but just want to know when HP changes and by how much. An example use case is a UI health bar that needs to update whenever HP changes, regardless of the source of the change

After the healing pipeline completes and the resurrection game action has been dispatched:

- **Entity Resurrected Event** — the dedicated resurrection signal, carrying a `ResurrectionContext` with the following fields:
 - **Target:** the `EntityCore` of the resurrected entity (convenience accessor for `ReceivedHeal.Target`)
 - **PreviousValue:** HP immediately before resurrection (at or below the death threshold)

- **NewValue:** HP after resurrection, computed as `PreviousValue + ReceivedHeal.HealAmount.NetAmount`
- **AbsAmount:** `NewValue - PreviousValue`
- **ReceivedHeal:** the full `ReceivedHealContext` produced by the internal `Heal` call, exposing:
 - **HealAmount:** a `HealAmountContext` with `RawAmount` (base amount before modifiers), `AfterModifierAmount` (after flat and percentage modifiers), and `NetAmount` (final HP actually added, capped to max HP)
 - **HealSource:** the `HealSourceSO` used for this resurrection
 - **Source:** the healer `EntityCore` (`null` for system-initiated resurrections, such as those triggered via the convenience overloads or the `ResurrectComponentGameActionSO`)
 - **IsCritical:** whether the resurrection heal was flagged as a critical

Each event is available in both global and extra variants on `EntityHealth`. Use global events when any `GameObject` in the game needs to receive the signal; use extra events for targeted, component-specific or GO-hierarchy-specific communications. See [Global Events](#) for guidance on choosing between them.

Limitations

Requirements

- Astra RPG Framework
- Unity 6 or later

Package Contents

Samples

No More Utils Folder

Diversamente da Astra RPG Framework, in questo package non troverete una cartella **Utils** tra i vari samples. Ho preso questa decisione perche' ritengo che sia piu' saggio e robusto lasciare che siate voi crearvi le vostre istanze. Questo perche' se costruite il vostro progetto attorno alle istanze che vi fornisco nei samples, e poi decidete di importare una nuova versione del package e reimportare i samples, rischiate di fare confusione su quali istanze di quale versione state attualmente usando per il vostro progetto. Lasciandovi creare a voi tutte le istanze degli oggetti del framework questa problematica viene completamente evitata.

Inoltre, le istanze presenti nei samples della scena di esempio sono marchiati con un suffisso (**Astra Health Samples**) che li rende facilmente identificabili e distinguibili da eventuali istanze che potreste creare voi per il vostro progetto.

Se volete prendere ispirazione dalle istanze presenti nei samples, potete farlo tranquillamente, ma vi consiglio di creare nuove istanze per il vostro progetto, in modo da evitare qualsiasi confusione futura.

Examples Folder - Overview

All'interno della cartella **Examples** troverete una scena di esempio: e' il vostro sandbox per provare le funzionalita' di Astra RPG Health. Rispetto a quella minimale di Astra RPG Framework, questa scena ha piu' contenuti e funzionalita' da esplorare. La scena copre tutte le funzionalita' principali di Astra RPG Health:

- Health System
- Damage Types
- Damage Sources
- Damage Reduction
- Defense Penetration
- Healing System
- Heal and Damage Modifiers
- Passive Regeneration
- Resurrection System
- Event System
- Health Scaling Component
- Experience Collection System

The Scene - Overview



Sample scene as you enter in play mode

Se aprite e avviate la scena di esempio, vedrete un combattente stilizzato sulla sinistra (Duelist) e un manichino sulla destra (Dummy). Senza troppe sorprese, potrete scatenare l'ira del vostro personaggio contro il povero manichino al fine di testare le funzionalita' di Astra RPG Health. Prima di prendercela con il manichino, diamo un'occhiata al resto della scena. Sopra il duelist c'e' un pannello contenente il nome del personaggio, il livello, l'eventuale classe, l'health bar, e infine i valori delle statistiche del personaggio. Un analogo pannello e' presente sopra il manichino. Potete scorrere con la rotellina del mouse su questi pannelli per visualizzare tutte le statistiche. Noterete che a differenza della scena di esempio di Astra RPG Framework, in questa scena i pannelli non mostrano gli attributi. Questo perche' ho volontariamente escluso il loro utilizzo in questa scena, al fine di concentrarci sulle funzionalita' di Astra RPG Health. Se avessi incluso anche gli attributi, sarebbe stato piu' complesso ragionare ragionare sull'effetto delle formule di calcolo del danno, delle riduzioni, delle penetrazioni e cosi' via, dal momento che i valori delle statistiche sarebbero stati influenzati anche dagli attributi. Quindi, per semplicita', in questa scena di esempio dovrete concentrarvi solo sulle statistiche.

Oltre ai pannelli delle statistiche, noterete che sono presenti anche dei pulsanti con una o due icone alla loro sinistra. Questi rappresentano le abilita' che ogni personaggio puo' utilizzare. E si', anche il dummy in questo caso ha voce in capitolo e puo' vendicarsi dei sopprusi subiti nei vari videogiochi negli ultimi decenni.

Ogni personaggio ha 3 abilita'; ogni riga di icona/e + 2 pulsanti costituisce una abilita' a se'. Per ciascuna abilita', il primo pulsante ne contiene il nome ed e' il punto d'attivazione. Il pulsante alla sua destra e' un toggle che definisce se l'abilita' effettuera' un colpo critico o meno. Infine, l'icona, o le icone, alla sinistra del nome dell'abilita' rappresentano i tipi di danno che i vari effetti dell'abilita' possono infliggere, o una

cura se l'effetto dell'abilita' e' di guarigione. Una singola icona significa che l'abilita' ha un singolo effetto, mentre due icone indicano che l'abilita' ha due o piu' effetti. Nel caso vi fossero, 3 o piu' effetti, solo le icone dei primi 2 effetti saranno mostrate. Le icone che troverete saranno in tutto 4:

- Una spada arancione, che rappresenta il danno fisico
- Una fiamma viola, che rappresenta il danno magico
- Uno scudo bianco perforato, che rappresenta il danno puro
- Una croce verde, che rappresenta la guarigione

La spada, la fiamma e lo scudo rappresentano quindi effetti di danno, e sono associate ai rispettivi damage types. La croce verde rappresenta invece un effetto di guarigione.

⚠ **WARNING**

Ci tengo a specificare che l'implementazione data per le abilita' in questa scena di esempio e' semplicistica e non rappresenta un sistema di abilita' completo. Era necessario introdurre un ability system minimale per poter dimostrare le funzionalita' di Astra RPG Health, ma non considerate questa implementazione come un punto di riferimento per la creazione di un sistema di abilita' piu' complesso.

Sara' responsabilita' di un futuro package di estensione di Astra fornire un sistema di abilita' completo e flessibile. Per ora, considerate questa implementazione come un semplice strumento per testare le funzionalita' di Astra RPG Health.

Noterete inoltre che appena sopra la testa del duelist ci sono un paio di pulsanti, che recitano "Next Hero" e "Previous Hero". Questi pulsanti vi permettono di cambiare il vostro personaggio. Oltre al duelist, che si focalizza su abilita' che arrecano danno fisico e di guarigione, ci sono altri 2 personaggi, ognuno con un focus diverso. Il secondo personaggio, l'Assassin, si focalizza su abilita' che arrecano danno fisico e puro, talvolta con scaling sulla salute del nemico. Il terzo personaggio, il Mage, si focalizza su abilita' che arrecano danno magico.

The Hierarchy - Overview

Se date ora un'occhiata alla gerarchia della scena vedrete che, oltre gli oggetti principali di default, contiene:

- l'entita' dummy
- il canvas del dummy (pannello delle statistiche e pannello delle abilita')
- una sezione con le entita' dei tre personaggi giocabili (duelist, assassin e mage). Il duelist e' attivo all'avvio della scena, mentre gli altri due personaggi sono disattivati. Cambiando personaggio con i pulsanti "Next Hero" e "Previous Hero", ciclerete tra questi tre personaggi, attivandone uno alla volta e disattivando gli altri due.

- una sezione con i canvas dei tre personaggi giocabili (pannello delle statistiche e pannello delle abilita' per ciascun personaggio). Anche in questo caso, il canvas del duelist e' attivo all'avvio della scena, mentre gli altri due canvas sono disattivati.
- Un oggetto **HeroSelector**. Questo oggetto ha come figli i game object che rappresentano i pulsanti "Next Hero" e "Previous Hero".
- Un oggetto **PopupCanvas**, che contiene due sotto-oggetti: **DamagePopupManager** e **HealPopupManager**. Questi oggetti sono responsabili della creazione dei popup di danno e guarigione che appaiono sopra la testa dei personaggi quando subiscono danno o ricevono guarigione.

Ciascuna entita' ha i componenti Astra **EntityCore, **EntityClass**, **EntityStats**, e il nuovo **EntityHealth** nell'oggetto radicale.** Il manichino non possiede una classe. Questa scelta non e' per discriminare il nostro amico legnoso, ma semplicemente perche' semplifica il cambiamento delle statistiche del manichino al fine di agevolare il testing delle funzionalita' del package. In questo modo, se voleste cambiare l'Armor per vedere come cambia la riduzione del danno in real-time, non dovete andare ad operare attraverso la Growth Formula associata a quella statistica nella classe del manichino, ma potete semplicemente modificarne il valore attraverso **EntityStats**. Semplice ed efficace.

Ogni entita' (compreso il dummy) ha inoltre due sotto-oggetti: **Visuals** e **Skills**. Il primo contiene tutti gli oggetti responsabili della rappresentazione visiva del personaggio (sprite e health bar per il dummy, sprite e sprite dell'attacco per i personaggi del giocatore), mentre il secondo serve per specificare le abilita' del personaggio.

Gli oggetti relativi ai canvas delle entita' hanno invece i seguenti sotto-oggetti: un **HeroPanel** responsabile di mostrare il pannello con nome, livello, classe, health bar e statistiche del personaggio, e uno **SkillsPanel**, che contiene i pulsanti e le icone relativi alle abilita' del personaggio.

The Project Files - Overview

Nell'esploratore delle risorse, nei samples del package, oltre la scena di esempio, avrete queste cartelle:

- **Art:** Contiene tutte le risorse artistiche utilizzate nella scena di esempio.
- **Instances:** Questa e' la cartella piu' importante, in quanto contiene tutte le istanze degli oggetti forniti da Astra RPG Health. Vi ritroverete a interagire molto con queste istanze per testare le funzionalita' del package, e potrete prendere ispirazione da esse per creare le vostre istanze personalizzate per il vostro progetto.
- **Prefabs:** Contiene i prefab di tutti gli oggetti presenti nella scena di esempio.
- **Resources:** Contiene l'istanza di configurazione di Astra RPG Health per la scena di esempio.
- **Scripts:** Contiene tutti gli script utilizzati nella scena di esempio.

⚠ WARNING

Una nota in merito a **Resources**. Come spiegato in [Global Settings](#), Astra RPG Health utilizza un'istanza di **AstraRpgHealthConfigSO** per configurare le sue funzionalita' nel vostro progetto Unity. Questa specifica istanza fornita nei samples e' stata configurata per la scena di esempio. La scena funziona sin da subito perche la risorsa di configurazione, chiamandosi esattamente **Astra Rpg Health Config**, viene automaticamente caricata da Astra RPG Health all'import del package e assegnata nella configurazione globale del package.

Quando andrete a creare la vostra istanza **AstraRpgHealthConfigSO** di configurazione per il vostro progetto, dovrete assegnarla manualmente alla configurazione globale di Astra RPG Health, come spiegato in [Project Settings](#) attraverso le project settings, o in alternativa, come spiegato in [Manual Configuration](#), assegnandola direttamente all'istanza **AstraRpgHealthGlobalSettings** presente nella cartella **Assets/Resources** (questa cartella resources e' collocata radicalmente negli Assets, non nei samples del package).

Se non assegnate la vostra istanza di configurazione alla configurazione globale di Astra RPG Health, il package continuera' ad utilizzare la configurazione dei samples, creando confusione e potenziali problemi di configurazione.

Instances Folder

Ora ci concentreremo sulla cartella **Instances**, che e' quella piu' rilevante per voi.

Spare objects: Default Damage Calculation Strategy and Lifesteal Config

Partiamo da due oggetti sfusi: **Default Damage Calculation Strategy** e **Lifesteal Config**. Il primo rappresenta la [strategia di calcolo del danno](#) utilizzata di default nella scena. Il secondo rappresenta la [configurazione del lifesteal](#); tutti i tipi di danno utilizzano la stessa stat **Lifesteal** per il calcolo del lifesteal.

Statistics and Stat Sets

Nella cartella **Stats** trovate tutte le istanze delle **Stat** e degli **StatSet** utilizzati nella scena di esempio. Potete esplorare voi nel dettaglio le statistiche e gli stat set, l'unica cosa che sottolineo e' che l'**Hero Stat Set** e' lo stat set utilizzato da tutti e 4 i personaggi della scena di esempio (duelist, assassin, mage e dummy).

Classes

Nella sottocartella **Classes** trovate tutte le classi per i 3 personaggi giocabili e tutte le relative growth formula che definiscono la progressione delle loro statistiche. C'e' anche una cartella **Common** che contiene alcune utilita' condivise da piu' classi. Ad esempio, la **200% Const Critical Multiplier GF** e' una growth

formula che restituisce un moltiplicatore di 200% per i colpi critici a tutti i livelli. Solitamente un gioco utilizza un moltiplicatore fisso per tutte le entita' a tutti i livelli, quindi non ha senso duplicare questa growth formula in ogni classe, ma e' piu' efficiente creare un'unica istanza singola e condividerla tra tutte le classi che ne hanno bisogno.

Damage Sources

Nella sottocartella **Damage Sources** trovate tutte due damage sources: **Skill** e **Environmental**. La scena non fa uso di danni ambientali, ma la damage source e' utilizzata nella [Experience Collection](#), in particolare nella strategia **Environmental Kill Exp Strategy**. Quindi, sebbene non sia testabile nella scena, ha valore di esempio per mostrarvi come creare e configurare una strategia multipla che usa sia la direct kill che la environmental kill strategies.

Damage Types

Nella sottocartella **Damage Types** trovate i tre tipi di danno utilizzati nella scena di esempio: Physical, Magical e True, e due cartelle contenenti le tre varianti di Damage Reduction Functions e Defense Reduction Functions. Attualmente, sia physical che magical damage utilizzano la formula logaritmica per la riduzione del danno sulla base della statistica difensiva, e la riduzione percentuale per la penetrazione della statistica difensiva. Le ho fornite tutte e tre per agevolarvi il testing qualora voleste sostituire le formule al volo.

Events

In **Events** invece trovate tutte le istanze degli eventi di gioco utilizzate dalla scena di esempio. Tutte le istanze appartengono ad Astra RPG Health eccetto per **Entity Levelled Up Game Event** e **Entity Levelled Down Game Event**. Questi eventi sono stati collegati come [global events](#) nella configurazione **AstraRpgHealthConfigSO**.

Tutte le comunicazioni tra i vari oggetti della scena di esempio vengono gestite attraverso questi eventi.

Experience Collection

Nella cartella **Experience Collection** ci sono tutte le istanze e le strategie utilizzate per la raccolta dell'esperienza nella scena di esempio. Come menzionato in precedenza, la scena di esempio non prevede l'utilizzo di danni ambientali, ma ho comunque incluso una strategia **Environmental Kill Exp Strategy** per mostrarvi come creare e configurare una strategia multipla che utilizza sia la direct kill che la environmental kill strategies.

Heal Sources

A differenza delle damage sources, nella scena di esempio vengono usate tutte e 4 le **HealSourceSO** che trovate nella cartella **Heal Sources**:

- **HP Regeneration**: utilizzata dagli eventi di rigenerazione passiva della salute

- **Lifesteal**: utilizzata dagli effetti di lifesteal
- **Resurrection**: utilizzata dalla cura applicata alla resurrezione di un'entita'
- **Skill**: utilizzata dalle abilita' che hanno effetti di guarigione

On Death Game Actions & On Resurrection Game Actions

In **On Death Game Actions** e **On Resurrection Game Actions** trovate invece le istanze delle game actions utilizzate per definire il comportamento delle entita' in risposta alla loro morte e resurrezione rispettivamente. Tratteremo meglio questo argomento piu' tardi in questa pagina.

Skills

Infine, nella cartella **Skills** trovate tutte le istanze delle abilita' utilizzate nella scena di esempio, suddivise per personaggio. Ogni abilita' ha una propria istanza di **SkillSO**, e una **ScalingFormula** con uno o piu' **ScalingComponent** associati. Alcune abilita', come menzionato prima, possono avere piu' di un effetto. In tal caso, hanno una scaling formula per ogni effetto.

Interacting with the Scene

Ora che abbiamo esplorato la scena di esempio, la gerarchia e i file di progetto, vi do alcune informazioni su alcune configurazioni specifiche che meritano due parole in piu'.

Casting skills to deal damage and heal

Le skill della sample scene si possono riassumere, dal punto di vista tecnico, come dei costruttori di **PreDamageContext** e **PreHealContext** che instradano questi contesti verso le entita' bersaglio. Il bersaglio processerà questi contesti nei suoi metodi **heal** e **TakeDamage**. Il calcolo dei danni che avviene all'interno di **TakeDamage**, come abbiamo visto nei workflows, passa attraverso la pipeline del calcolo del danno. La pipeline usa la strategia che è stata configurata a livello di configurazione **AstraRPGHealthConfigSO**. Quando l'entità viene curata o subisce danni, invocherà i rispettivi Global Events configurati anch'essi nella configurazione del package. E gli heal e damage pop-up managers che trovate nella hierarchy sono in ascolto di questi due eventi rispettivamente. Quando ricevono l'evento, creano un pop-up sopra la testa del personaggio bersaglio che mostra l'ammontare di danno subito o guarigione ricevuta. I pop up del danno avranno colori e icone diverse in base al tipo di danno inferto. Le icone (e il colore) combaceranno con quelle che vedete alla sinistra dell'abilità che avete lanciato.

Installation instructions

Importing Astra RPG Health and its samples

1. From the package manager, import Astra RPG Health. You can find the package in the "My Assets" section.
2. After importing, head to the "In Project" section, always in the Package Manager, and click on "AstraRPGHealth".
3. Click on the "Samples" tab and import the samples you desire. Find out more about the samples in the [Samples documentation](#).
4. If you imported the "Example scene and instances" samples you need to import also TextMeshPro Essentials. Click on "Window > TextMeshPro > Import TMP Essential Resources".

Changelog

All notable changes to this project will be documented in this file.

The format is based on [Keep a Changelog](#) .

[1.0.0] - Unreleased

Added

- Initial release of Astra RPG Health.

Migration Guide

Any breaking changes in Astra RPG Health will be documented here, along with instructions on how to update your project to be compatible with the latest version of the package.